Geophysical Journal International

Geophys. J. Int. (2012) **191,** 813–831 doi: 10.1111/j.1365-246X.2012.05658.x

Angle between principal axis triples

Walter Tape[1] and Carl Tape[2]

[1]*Department of Mathematics, University of Alaska, Fairbanks, Alaska* 99775*, USA*

[2]*Geophysical Institute and Department of Geology & Geophysics, University of Alaska, Fairbanks, Alaska* 99775*, USA. E-mail: carltape@gi.alaska.edu*

Accepted 2012 August 21. Received 2012 August 19; in original form 2012 May 14

SUMMARY

The principal axis angle ξ_0, or Kagan angle, is a measure of the difference between the orientations of two seismic moment tensors. It is the smallest angle needed to rotate the principal axes of one moment tensor to the corresponding principal axes of the other. This paper is a conceptual review of the main features of ξ_0. We give a concise formula for calculating ξ_0, but our main goal is to illustrate the behaviour of ξ_0 geometrically. When the first of two moment tensors is fixed, the angle ξ_0 between them becomes a function on the unit ball. The level surfaces of ξ_0 can then be depicted in the unit ball, and they give insights into ξ_0 that are not obvious from calculations alone. We also include a derivation of the known probability density p_{ξ_0} of ξ_0. The density $p_{\xi_0}(t)$ is proportional to the area of a certain surface $\mathbb{L}_w(t)$. The easily seen variation of $\mathbb{L}_w(t)$ with t then explains the rather peculiar shape of p_{ξ_0}. Because the curve p_{ξ_0} is highly non-uniform, its shape needs to be considered when analysing distributions of empirical ξ_0 values. We recall an example of Willemann which shows that ξ_0 may not always be the most appropriate measure of separation for moment tensor orientations, and we offer an alternative measure.

Key words: Theoretical seismology.

1 INTRODUCTION

The seismic moment tensor is a mathematical expression of an earthquake source. In matrix terms, it is a 3 × 3 symmetric matrix; its eigenvalues express the size and type of the earthquake source, and its eigenvectors express the orientation. When comparing moment tensors, some quantitative measure of separation between them is often desirable. A natural measure is the ordinary euclidean distance in the space of 3 × 3 matrices. But while the euclidean distance is the best overall measure, it leaves something to be desired conceptually, since it does not reveal the separate contributions of the size, type, and orientation. Some supplementary measures are therefore desirable as well. Differences in size and type, that is, differences in eigenvalue triples, are relatively easy to measure. It is less obvious what should constitute a reasonable measure of difference in moment tensor orientation.

In this paper we review one commonly used measure, which we call the principal axis angle and denote by ξ_0. The principal axis angle ξ_0 between two moment tensors is the smallest angle required to rotate the principal axes of one of them into the corresponding principal axes of the other. In Section 3 we give a more precise definition of ξ_0 as well as a concise formula for its calculation (eq. 34). In Sections 4 and 7 we give some intuition for ξ_0 by depicting it as a function on the unit ball. In Section 5 and Appendix A we give a derivation of the known probability density p_{ξ_0} for ξ_0, the underlying assumption being that the moment tensor orientations are random. We give a geometric characterization of p_{ξ_0} that helps to explain its distinctive and highly non-uniform nature. We remind readers that the distinctive shape of p_{ξ_0} should be kept in mind when interpreting ξ_0 values for real data sets. In Section 8 we suggest that ξ_0 may not always be the best measure of separation for moment tensor orientations, and we offer an alternative measure.

Although we have just described ξ_0 as a measure of difference in moment tensor orientation, the only relevant feature of the moment tensor is its orthorhombic symmetry, which is embodied in its principal axes. Orthorhombic symmetry is the symmetry of a brick, or of a three-coloured cube whose opposite faces are coloured the same. The change in orientation of any object having orthorhombic symmetry can be measured by ξ_0.

The angle ξ_0 has probably seen its widest application and most thorough study in connection with crystalline textures, where it is just one of many 'misorientation angles' each associated with a symmetry group (e.g. Sutton & Balluffi 1995; Morawiec 2004). Mackenzie & Thomson (1957) introduced the concept of misorientation angle in connection with cubic symmetry and used Monte Carlo methods to approximate its probability density function p_{ξ_0}. Mackenzie (1958) and Handscomb (1958) each gave an analytic expression for p_{ξ_0}, again for cubic symmetry. Grimmer (1979a,b) found p_{ξ_0} for other symmetry groups. Grimmer gave no derivation but said that he had used the methods of Handscomb (1958). Later Morawiec (1995), using ideas of Frank (1988) as well as

Handscomb (1958), published a derivation of p_{ξ_0} and gave examples for many symmetry groups.

In seismology the angle ξ_0 has been studied by Kagan (e.g. 1991, 2007), and in fact in some literature the angle is referred to as the Kagan angle. The angle ξ_0 has been used in the statistical analysis of real and synthetic sets of moment tensors, faults, and stress (e.g. Kagan 1992; Hardebeck & Shearer 2002; Hardebeck 2006). Hardebeck, for example, was able to infer unexpected homogeneity in earthquake faulting in California. The angle has also been used to compare moment tensors from different earthquakes (e.g. Okal *et al.* 2011). And it has been used in comparing moment tensors computed for the same events using different techniques (Frohlich & Davis 1999; Pondrelli *et al.* 2006; Nakano *et al.* 2010; Konstantinou & Rontogianni 2011; Duputel *et al.* 2012; Imprescia *et al.* 2012).

As methods of measurement and modeling continue to improve (e.g. Liu *et al.* 2004), the principal axis angle ξ_0 is apt to find increased use in measuring moment tensor differences. In spite of its simple description as a minimum rotation angle, the angle ξ_0 has some subtle features that users should be aware of. The purpose of this paper is to provide a conceptual exposition of ξ_0 that will elucidate those features.

2 ROTATION MATRICES

This section reviews some prerequisites involving rotation matrices. Most of the results are stated without proof. *Wikipedia* is a sufficient reference.

2.1 Rotation axis and rotation angle

A rotation matrix is a 3×3 orthogonal matrix U with determinant equal to 1. That is,

$$UU^{\mathrm{T}} = I, \tag{1a}$$

$$\det U = 1, \tag{1b}$$

where U^{T} is the transpose of U and where I is the identity matrix. If U is considered as a linear transformation of $\mathbb{R}^3$, then eq. (1a) says that U preserves distances, and eq. (1b) says that U preserves handedness.

A typical rotation matrix U has two normalized rotation axes and two rotation angles. If ξ is the rotation angle associated with the rotation axis $\mathbf{u}$, then the effect of U is to rotate each point in $\mathbb{R}^3$ through angle ξ about the axis $\mathbf{u}$, with positive rotation being counterclockwise when $\mathbf{u}$ determines the 'up' direction. If ξ is the rotation angle associated with the rotation axis $\mathbf{u}$, then $-\xi$ is the rotation angle associated with the rotation axis $-\mathbf{u}$.

We will always choose the rotation axis $\mathbf{u}$ and associated rotation angle ξ so that $0 \le \xi \le \pi$. Then, $\mathbf{u}$ (normalized) and ξ satisfy

$$\cos\xi = \frac{-1 + \operatorname{trace} U}{2}, \quad 0 \le \xi \le \pi, \tag{2a}$$

$$U\mathbf{u} = \mathbf{u}, \quad \|\mathbf{u}\| = 1, \tag{2b}$$

$$(U\mathbf{e}) \cdot (\mathbf{u} \times \mathbf{e}) \ge 0, \tag{2c}$$

where $\mathbf{e}$ is a non-zero vector orthogonal to $\mathbf{u}$. Eq. (2a) uniquely determines ξ. If U is neither the identity nor a 180° rotation, then eqs (2b) and (2c) uniquely determine $\mathbf{u}$; eq. (2b) determines $\mathbf{u}$ up to sign, and then eq. (2c) chooses between the two sign possibilities, by requiring that the triple $\mathbf{u}$, $\mathbf{e}$, $U\mathbf{e}$ be right-handed. If U is a 180° rotation, then it has two normalized rotation axes $\pm\mathbf{u}$, both satisfying eqs (2). (The left-hand side of eq. (2c) is then zero.) And if U is the identity, then any unit vector $\mathbf{u}$ satisfies eqs (2) with $\xi = 0$.

Eqs (2b) and (2c) nicely express the conceptual meaning of the rotation axis, but they have the disadvantage that they require U to be truly orthogonal, not just approximately so. Appendix F has another approach to calculating the rotation axis.

2.2 Quaternion summary

A unit quaternion has the form

$$q = (w, x, y, z), \qquad w^2 + x^2 + y^2 + z^2 = 1, \tag{3}$$

where w, x, y, z are real numbers. We also use the notations

$$\begin{aligned} q &= (w, \mathbf{v}) = (w,\ (x, y, z)), \\ q &= w + xi + yj + zk. \end{aligned} \tag{4}$$

To multiply quaternions, one writes them in the latter form and then proceeds as one might hope, except that the multiplication is not commutative, and

$$i^2 = j^2 = k^2 = ijk = -1. \tag{5}$$

For unit quaternions q, the inverse is the conjugate. That is,

$$q^{-1} = \overline{q}, \tag{6}$$

where $\overline{(w, \mathbf{v})} = (w, -\mathbf{v})$.

There is a two-to-one correspondence between unit quaternions and rotation matrices. The rotation matrix associated with the unit quaternion $q = (w, x, y, z)$ is

$$U(q) = \begin{pmatrix} w^2 + x^2 - y^2 - z^2 & 2(xy - wz) & 2(wy + xz) \\ 2(xy + wz) & w^2 - x^2 + y^2 - z^2 & 2(yz - wx) \\ 2(xz - wy) & 2(wx + yz) & w^2 - x^2 - y^2 + z^2 \end{pmatrix}. \tag{7}$$

From eq. (7),

$$\begin{aligned} U(q_2)\,U(q_1) &= U(q_2\,q_1), \\ U(-q) &= U(q). \end{aligned} \tag{8}$$

The unit quaternions for a rotation U are $\pm q$, where

$$q = \left(\cos\frac{\xi}{2},\ \left(\sin\frac{\xi}{2}\right)\mathbf{u}\right), \tag{9}$$

and where ξ and $\mathbf{u}$ are the rotation angle and normalized rotation axis for U. The rotation angle for a rotation U is therefore given by

$$\cos\frac{\xi}{2} = |w|, \tag{10}$$

where (w, x, y, z) is a unit quaternion for U.

The 180° rotations X_π, Y_π, Z_π about the x, y, z coordinate axes play an important role in what follows (e.g. Fig. 3). From eq. (7), their respective unit quaternions are $\pm i$, $\pm j$, and $\pm k$. If $q = w + xi + yj + zk$ is a unit quaternion for U, then unit quaternions for U, UX_π, UY_π, UZ_π are, respectively,

$$q = w + xi + yj + zk, \tag{11a}$$

$$qi = (w + xi + yj + zk)i = -x + wi + zj - yk, \tag{11b}$$

$$qj = (w + xi + yj + zk)j = -y - zi + wj + xk, \tag{11c}$$

$$qk = (w + xi + yj + zk)k = -z + yi - xj + wk. \tag{11d}$$

From eq. (10) the rotation angles ξ, ξ_x, ξ_y, ξ_z for U, UX_π, UY_π, UZ_π are therefore given by

$$\begin{aligned} \cos\frac{\xi}{2} &= |w|, \\ \cos\frac{\xi_x}{2} &= |x|, \\ \cos\frac{\xi_y}{2} &= |y|, \\ \cos\frac{\xi_z}{2} &= |z|. \end{aligned} \tag{12}$$

Thus the entries of the unit quaternion for U are the rotation angles, thinly disguised, of U, UX_π, UY_π, UZ_π.

2.3 Depicting rotations in the unit ball

Eq. (9) suggests that rotation matrices be represented as points in the unit ball, that is, in the set

$$\mathbb{B} = \{\mathbf{v} \in \mathbb{R}^3 : \|\mathbf{v}\| \le 1\}. \tag{13}$$

The correspondence between rotations U and points $\mathbf{v}$ of $\mathbb{B}$ is

$$U \longleftrightarrow \mathbf{v} = \left(\sin\frac{\xi}{2}\right)\mathbf{u}, \tag{14}$$

where ξ and $\mathbf{u}$ are the rotation angle and normalized rotation axis of U. Thus each point $\mathbf{v}$ in $\mathbb{B}$ determines a rotation matrix; the direction of $\mathbf{v}$ gives the rotation axis, and the norm of $\mathbf{v}$ determines the rotation angle ξ by

$$\|\mathbf{v}\| = \sin\frac{\xi}{2}. \tag{15}$$

Stated otherwise, the rotation matrix at the point $\mathbf{v}$ is $U(q)$, where $U(q)$ is as in eq. (7) and where q is the unit quaternion $(\sqrt{1 - \|\mathbf{v}\|^2},\ \mathbf{v})$. See Fig. 1.

Thus the unit ball $\mathbb{B}$ represents the group $\mathbb{U}$ of all rotations. All rotations can in principle be visualized at once, as suggested in Fig. 2. First some standard object is chosen, such as the cube in the figure; this is the 'initial object'. Then, for each point $\mathbf{v}$ in the unit ball, the corresponding rotation U is applied to the initial object, and the result is displayed at $\mathbf{v}$. Since the rotation corresponding to the zero vector is the identity, the object seen at the origin is necessarily the initial object itself. By comparing the object at $\mathbf{v}$ with the object at the origin, one sees the effect of the rotation U.

Points $\mathbf{v}$ on the unit sphere—the boundary of $\mathbb{B}$—have $\|\mathbf{v}\| = 1$ and hence have $\xi = \pi$ (eq. 15); they represent 180° rotations. Antipodal points on the unit sphere therefore represent the same rotation matrix. The points need to be considered the same when $\mathbb{B}$ represents rotations.

Usually we will not distinguish carefully between the unit ball $\mathbb{B}$ and the group $\mathbb{U}$ of rotations. Likewise, we will not distinguish between a point $\mathbf{v}$ of $\mathbb{B}$ and its corresponding rotation U in $\mathbb{U}$.

3 PRINCIPAL AXIS ANGLE ξ_0

A 'frame' is a right-handed orthonormal basis for $\mathbb{R}^3$. Initially the principal axis angle ξ_0 will be defined for pairs of frames. Eventually ξ_0 will be defined for pairs of moment tensors, by applying it to the frames associated with the principal axes of the moment tensors.

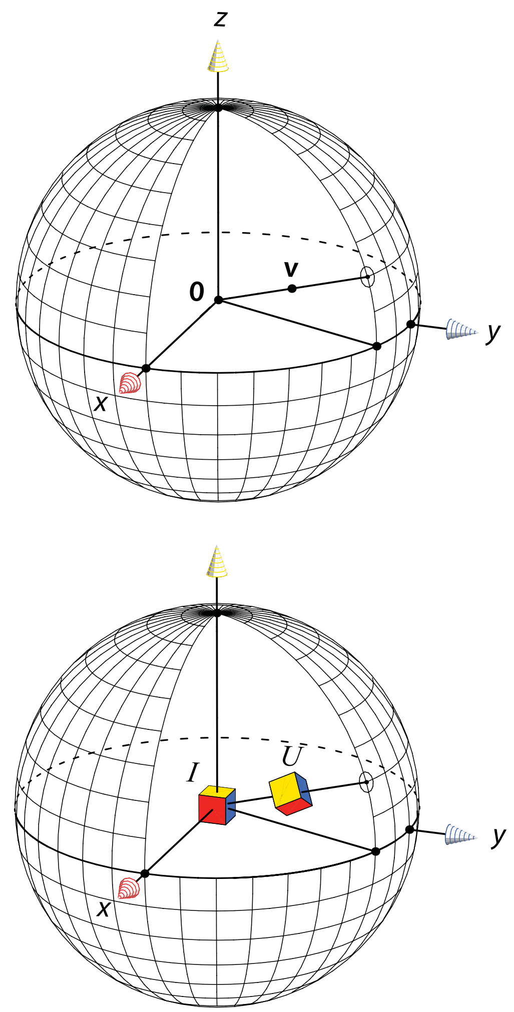

Figure 1. Rotation U determined by the point $\mathbf{v}$ in the unit ball. The cube at $\mathbf{v}$ is the result of applying U to the 'initial cube' at the origin $\mathbf{0}$; hence the cube at $\mathbf{v}$ represents U. The direction of $\mathbf{v}$ gives the rotation axis of U, and the length of $\mathbf{v}$ gives the rotation angle ξ by $\sin(\xi/2) = \|\mathbf{v}\|$. Here $\|\mathbf{v}\| = 0.5$ and so $\xi = \pi/3$.

In contrast to the frame vectors, the principal axes of a moment tensor are considered to be undirected lines rather than vectors. We will refer to any ordered triple of mutually perpendicular lines as a principal axis triple; the lines need not be associated with any moment tensor.

3.1 Distance between frames

We will usually treat frames and rotation matrices interchangeably; the columns of a rotation matrix are frame vectors and vice versa. Suppose then that the matrices U_1 and U_2 are frames. If $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ are the standard basis vectors, then $U_1\mathbf{e}_1$, $U_1\mathbf{e}_2$, $U_1\mathbf{e}_3$ (the columns of U_1) are the frame vectors of U_1. Likewise, $U_2\mathbf{e}_1$, $U_2\mathbf{e}_2$, $U_2\mathbf{e}_3$ are the frame vectors of U_2. Since

$$U_2U_1^{-1}(U_1\mathbf{e}_i) = U_2\mathbf{e}_i, \tag{16}$$

then $U_2U_1^{-1}$ is the rotation that takes frame U_1 to frame U_2, that is,

$$U_2U_1^{-1} : U_1 \to U_2. \tag{17}$$

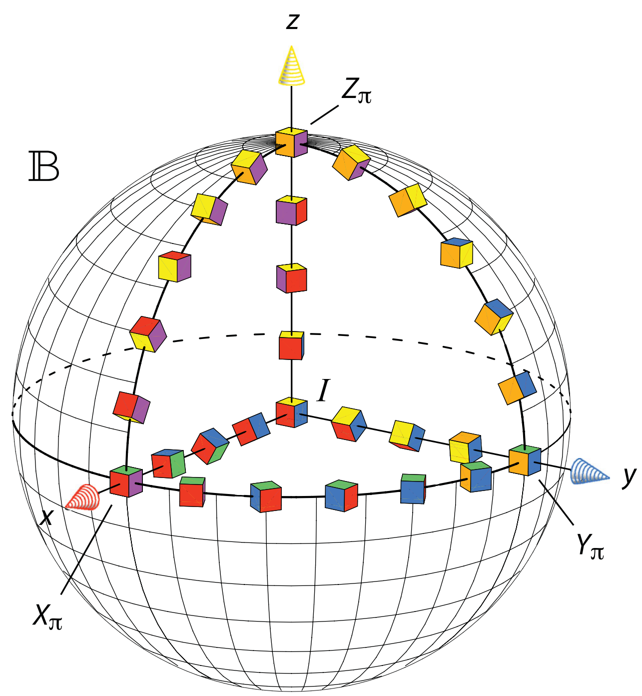

Figure 2. A suggestion of all possible rotations. At each point **v** of the unit ball $\mathbb{B}$, there is a corresponding rotation U and, in principle, a cube which is the result of applying U to the initial cube. The identity I is at the centre of the unit ball, and 180° rotations are on the unit sphere $\|\mathbf{v}\| = 1$. The rotations at the intercepts of the coordinate axes with the unit sphere are the 180° rotations X_π, Y_π, Z_π about the x, y, z axes. Here, with only a sample of cubes shown, much is left to the imagination.

It is therefore reasonable to define the distance between frames U_1 and U_2 by

$$d(U_1, U_2) = \xi(U_2 U_1^{-1}), \tag{18}$$

where $\xi(V)$ denotes the rotation angle of V. Then the function d is indeed a metric in the usual sense. That is,

$$d(U_1, U_2) \geq 0, \tag{19a}$$

$$d(U_1, U_2) = 0 \text{ iff } U_1 = U_2, \tag{19b}$$

$$d(U_1, U_2) = d(U_2, U_1) \qquad \text{(symmetry)}, \tag{19c}$$

$$d(U_1, U_2) + d(U_2, U_3) \geq d(U_1, U_3) \quad \text{(triangle inequality)}. \tag{19d}$$

Eq. (19c) follows from eq. (2a) and from the fact that trace U = trace (U^{-1}) if U is a rotation. Eq. (19d) is proved in Appendix C. The metric d is both right and left invariant:

$$d(U_1 V, U_2 V) = d(U_1, U_2), \tag{20a}$$

$$d(V U_1, V U_2) = d(U_1, U_2), \tag{20b}$$

the latter equation following from trace $(V U V^{-1})$ = trace U. From eq. (18),

$$d(I, U) = \xi(U). \tag{21}$$

Since we usually consider frames to be points of the unit ball $\mathbb{B}$, we sometimes need to distinguish between the distance d and the ordinary euclidean distance in $\mathbb{B}$. In that case we will refer to d as the rotation metric, or the d-metric, or something similar.

Huynh (2009) has much more on metrics for rotations.

3.2 Definition of ξ_0

Let X_ξ, Y_ξ, Z_ξ be the rotations through angle ξ about the x, y, z coordinate axes. In particular, X_π, Y_π, Z_π are the 180° rotations about the coordinate axes.

For any frame U, the frame UX_π can be written $(UX_\pi U^{-1})U$. The frame UX_π is therefore the result of rotating the frame U through an angle of 180° about its own first frame vector. Equivalently, UX_π results from U by reversing the directions of its second and third frame vectors. Similarly for UY_π and UZ_π. See Fig. 3.

Thus, for any frame U, the frames U, UX_π, UY_π, UZ_π all correspond to the same ordered triple of principal axes. Technically, in fact, the principal axis triple determined by the frame U is the set

$$\overline{U} = U\,\mathbb{D}_2 = \{U, UX_\pi, UY_\pi, UZ_\pi\}, \tag{22}$$

where $\mathbb{D}_2$ is the symmetry group

$$\mathbb{D}_2 = \{I, X_\pi, Y_\pi, Z_\pi\}. \tag{23}$$

To define a reasonable measure of separation ξ_0 between principal axis triples that are associated with frames U_1 and U_2, we therefore need to consider not just the rotation

$$U_1 \to U_2, \tag{24}$$

but rather all of the rotations

$$U_1 R_1 \to U_2 R_2, \qquad R_1, R_2 \in \mathbb{D}_2. \tag{25}$$

These rotations are the matrices $U_2 R_2 (U_1 R_1)^{-1}$ (eq. 17). The principal axis angle $\xi_0(U_1, U_2)$ is defined to be the smallest of their rotation angles.

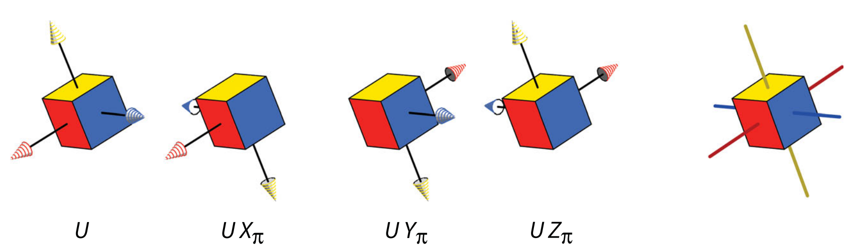

Figure 3. Frames (rotation matrices) U, UX_π, UY_π, UZ_π all corresponding to the same principal axis triple, at right. In each case the first, second, and third frame vectors are the red, blue, and yellow arrows, respectively. Any two of the frames differ by a 180° rotation about one of the three principal axes. Opposite faces of the cube are coloured the same, so the rotation of the cube itself cannot be detected here. When considered as rotation matrices, each of U, UX_π, UY_π, UZ_π has a rotation angle; the smallest of them is the principal axis angle $\xi_0(I, U)$.

Since $U_2R_2(U_1R_1)^{-1} = U_2R_2R_1U_1^{-1}$ and $R_2R_1 \in \mathbb{D}_2$, there are only four rotations to consider (not 16). Their rotation angles can be found from eq. (2a); the smallest of them gives ξ_0.

This brute force calculation of ξ_0 is workable, and it requires no calculation of rotation axes, but it does not give much insight. Eq. (34) below gives a more concise expression for ξ_0 that can be used to visualize ξ_0 as a function.

3.3 Some properties of ξ_0

In terms of the rotation distance d (eq. 18), the principal axis angle is

$$\xi_0(U_1, U_2) = \min_{R_1, R_2 \in \mathbb{D}_2} d(U_1R_1, U_2R_2). \tag{26}$$

For frames, the function ξ_0 fails to be a metric, because distinct frames can have zero separation ξ_0 between them. For example, $\xi_0(X_\pi, I) = 0$ but $X_\pi \neq I$. The remaining requirements for a metric, however, are satisfied. For $U_1, U_2, U_3 \in \mathbb{U}$,

$$\xi_0(U_1, U_2) \geq 0, \tag{27a}$$

$$\xi_0(U_1, U_2) = \xi_0(U_2, U_1) \qquad \text{(symmetry)}, \tag{27b}$$

$$\xi_0(U_1, U_2) + \xi_0(U_2, U_3) \geq \xi_0(U_1, U_3) \quad \text{(triangle inequality)}. \tag{27c}$$

Eqs (27a) and (27b) follow from the corresponding properties for d and from eq. (26). The triangle inequality for ξ_0 is proved in Appendix D.

From eqs (20b) and (26), the angle ξ_0 is left-invariant:

$$\xi_0(VU_1, VU_2) = \xi_0(U_1, U_2). \tag{28}$$

However, right invariance fails more often than not. As a reminder, we write

$$\xi_0(U_1V, U_2V) \stackrel{?}{=} \xi_0(U_1, U_2). \tag{29}$$

3.4 Distance between principal axis triples

From eq. (26),

$$\xi_0(U_1R_1, U_2R_2) = \xi_0(U_1, U_2), \qquad R_1, R_2 \in \mathbb{D}_2. \tag{30}$$

This is the fundamental property that allows ξ_0 to be defined not just for frames but for principal axis triples, and hence for moment tensors (Section 8). For principal axes triples $\overline{U_1}$ and $\overline{U_2}$ (eq. 22), the angle ξ_0 is defined by

$$\xi_0(\overline{U_1}, \overline{U_2}) = \xi_0(U_1, U_2). \tag{31}$$

Whether $\overline{U_1}$ is written as $\overline{U_1}$, $\overline{U_1X_\pi}$, $\overline{U_1Y_\pi}$, or $\overline{U_1Z_\pi}$, the result on the right-hand side of eq. (31) is the same, by eq. (30), and similarly for $\overline{U_2}$. But note that the analogue of eq. (31) for ξ would not work. This is the insight underlying the definition of ξ_0.

Eqs (27) and (28), which applied to frames, carry over automatically to principal axis triples, via eq. (31). Moreover,

$$\begin{aligned} \xi_0(\overline{U_1}, \overline{U_2}) = 0 &\Rightarrow \xi_0(U_1, U_2) = 0 \\ &\Rightarrow U_1 = U_2R \text{ for some } R \in \mathbb{D}_2 \\ &\Rightarrow \overline{U_1} = \overline{U_2}. \end{aligned} \tag{32}$$

On the set of principal axis triples, the function ξ_0 is therefore a metric. It is left-invariant.

The function d is a natural measure of distance between frames. The function ξ_0 is the associated distance between principal axis triples.

3.5 A more concise formula for ξ_0

Continuing from eq. (26),

$$\begin{aligned} \xi_0(U_1, U_2) &= \min_{R_1, R_2 \in \mathbb{D}_2} d(U_1R_1, U_2R_2) \\ &= \min_{R \in \mathbb{D}_2} d(I, U_1^{-1}U_2R) \\ &= \min(\xi, \xi_x, \xi_y, \xi_z), \end{aligned} \tag{33}$$

where $U = U_1^{-1}U_2$ and where ξ, ξ_x, ξ_y, ξ_z are the rotation angles for $U, UX_\pi, UY_\pi, UZ_\pi$, respectively. Then, from eqs (12), the principal axis angle $\xi_0(U_1, U_2)$ is given by

$$\cos\frac{\xi_0}{2} = \max(|w|, |x|, |y|, |z|), \tag{34}$$

where (w, x, y, z) is a unit quaternion for $U_1^{-1}U_2$. Such a quaternion is, from eq. (9),

$$\begin{aligned} w &= \cos(\xi/2), \\ (x, y, z) &= (\sin(\xi/2))\,\mathbf{u}, \end{aligned} \tag{35}$$

where ξ and $\mathbf{u}$ are the rotation angle and normalized rotation axis of $U_1^{-1}U_2$. The quaternion can also be calculated directly, as explained in Appendix F.

Appendix E has a sample calculation of ξ_0.

3.6 The case $U_1 = I$

From eq. (28) it follows immediately that

$$\xi_0(U_1, U_2) = \xi_0(I, U_1^{-1}U_2). \tag{36}$$

Thus the general case $\xi_0(U_1, U_2)$ can always be reduced to the case $\xi_0(I, U)$ in which the first frame is the identity. The matrix $U_1^{-1}U_2$ that appears in connection with eq. (34) is then just U. Stated less formally: one's own frame of reference can always be adjusted so that the first of U_1 and U_2 appears to be the identity frame.

In most of what follows, we will be interested in $\xi_0(I, U)$, which will then be abbreviated to $\xi_0(U)$.

3.7 The frame coordinates ξ, ξ_x, ξ_y, ξ_z

The frame coordinates ξ, ξ_x, ξ_y, ξ_z of a rotation U are defined to be the rotation angles of $U, UX_\pi, UY_\pi, UZ_\pi$, respectively:

$$\begin{aligned} \xi &= \xi(U), \qquad & \xi_x &= \xi(UX_\pi), \\ \xi_y &= \xi(UY_\pi), \qquad & \xi_z &= \xi(UZ_\pi). \end{aligned} \tag{37}$$

They are also the respective (rotation) distances from U to I, X_π, Y_π, Z_π. That is,

$$\begin{aligned} \xi &= \xi(U) = d(I, U), \\ \xi_x &= \xi(UX_\pi) = d(I, UX_\pi) = d(X_\pi, U), \\ \xi_y &= \text{etc.} \end{aligned} \tag{38}$$

The frame coordinates are related to the *xyz* cartesian coordinates in the unit ball $\mathbb{B}$ by eqs (12) and by

$$w = \left(1 - \|\mathbf{v}\|^2\right)^{1/2}, \quad \mathbf{v} = (x, y, z). \tag{39}$$

The four frame coordinates are not independent, however, since $w^2 + x^2 + y^2 + z^2 = 1$. Moreover, the eight symmetrically positioned points $(\pm x, \pm y, \pm z)$ all have the same frame coordinates, due to the absolute values in eqs (12), so the frame coordinates do not uniquely determine a frame unless an octant of $\mathbb{B}$ is specified.

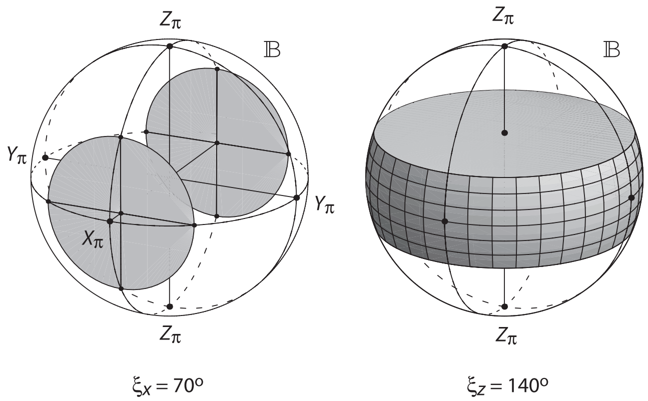

Figure 4. Two sets of frames. (*Left*) The set $\xi_x = t$ for $t = 70°$. The frame coordinates ξ, ξ_x, ξ_y, ξ_z for a frame U are the rotation angles of U, UX_π, UY_π, UZ_π, respectively. Hence the set $\xi_x = t$ consists of the frames U for which the rotation angle of UX_π is t. As a subset of the unit ball $\mathbb{B}$, the set $\xi_x = t$ is the pair of discs $|x| = \cos(t/2)$. (*Right*) The set $\xi_z \geq t$ for $t = 140°$. It consists of the frames U for which the rotation angle of UZ_π is at least t. In $\mathbb{B}$ it is the slab bounded by the horizontal discs $|z| = \cos(t/2)$.

In $\mathbb{B}$ the level surfaces for ξ are spheres, and the level surfaces for ξ_x, ξ_y, ξ_z are planes or, rather, discs. More precisely, the level surface $\xi = t$ is the sphere of euclidean radius $\sin(t/2)$, and the level surface $\xi_x = t$ is the pair of discs $|x| = \cos(t/2)$ perpendicular to the x-axis. The level surfaces $\xi_y = t$ and $\xi_z = t$ are analogous.

Many sets relevant to ξ_0 can be usefully described in terms of their frame coordinates. Consider, for example, the set

$$\{U \in \mathbb{U} : \xi_z(U) \geq t\},$$

which we abbreviate to

$$\{\xi_z \geq t\} \quad \text{or} \quad \xi_z \geq t.$$

It consists of the rotations U such that the rotation angle of UZ_π is at least t. Equivalently, it is the set of rotations whose rotation distance (eq. 18) to Z_π is at least t. In $\mathbb{B}$ it is the set $|z| \leq \cos(t/2)$, which is the horizontal slab between the two discs $|z| = \cos(t/2)$. Fig. 4 shows the slab for $t = 140°$. Table 1 gives several other sets of rotations, both in frame coordinates and in cartesian coordinates.

From eq. (33), the parameter $\xi_0(U) = \xi_0(I, U)$ is expressed in terms of the frame coordinates as

$$\xi_0 = \min(\xi,\ \xi_x,\ \xi_y,\ \xi_z). \tag{40}$$

Hence $\xi_0(U)$ is the (rotation) distance from U to the closest of I, X_π, Y_π, Z_π.

3.8 A partition of the unit ball

The group $\mathbb{U}$ of rotations, or frames, is not an efficient representation of principal axis triples, since each such triple corresponds to four frames (Fig. 3). There are many ways to partition $\mathbb{U}$, or the unit ball $\mathbb{B}$, into four subsets in such a way that each subset gives an efficient representation of principal axis triples.

Eq. (40) suggests that $\mathbb{B}$ be partitioned into subsets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ according to whether the principal axis angle ξ_0 is ξ, ξ_x, ξ_y, or ξ_z. In frame coordinates,

$$\begin{aligned} \mathbb{B}_w &= \{\xi_0 = \xi\}, \\ \mathbb{B}_x &= \{\xi_0 = \xi_x\},\ \mathbb{B}_y = \text{etc.}, \end{aligned} \tag{41}$$

or

$$\begin{aligned} \mathbb{B}_w &= \{\min(\xi,\ \xi_x,\ \xi_y,\ \xi_z) = \xi\}, \\ \mathbb{B}_x &= \{\min(\xi,\ \xi_x,\ \xi_y,\ \xi_z) = \xi_x\},\ \mathbb{B}_y = \text{etc.} \end{aligned} \tag{42}$$

The sets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ are separated from each other by the surfaces $\xi = \xi_x$, $\xi = \xi_y$, $\xi = \xi_z$, $\xi_x = \xi_y$, $\xi_y = \xi_z$, and $\xi_z = \xi_x$. The latter three are pairs of planes. The surface $\xi = \xi_x$ is (eqs 12)

$$\begin{aligned} \cos\frac{\xi}{2} &= \cos\frac{\xi_x}{2} \\ w &= |x| \\ 1 - x^2 - y^2 - z^2 &= x^2 \\ 2x^2 + y^2 + z^2 &= 1, \end{aligned} \tag{43}$$

which is an ellipsoid of revolution. It has vertices $(\pm 1/\sqrt{2}, 0, 0)$ on the x-axis, and its equatorial section is the unit circle $y^2 + z^2 = 1$ in the yz-plane. Points $\mathbf{v} = (x, y, z)$ that are on or inside the ellipsoid have $w \geq |x|$, that is, $\xi \leq \xi_x$. The ellipsoids $\xi = \xi_y$ and $\xi = \xi_z$ are analogous. Points $\mathbf{v}$ that are on or within all three ellipsoids have $\xi = \min(\xi, \xi_x, \xi_y, \xi_z)$; they are the points in $\mathbb{B}_w$. See Figs. 5 and 6.

The definitions of $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ can be paraphrased as follows. Each point $\mathbf{v} \in \mathbb{B}$ represents a frame U. Associated with U are the four frames U, UX_π, UY_π, UZ_π. If the smallest of the rotation angles for U, UX_π, UY_π, UZ_π is the rotation angle for U itself, then the point $\mathbf{v}$ is in $\mathbb{B}_w$ (green in Fig. 5). If the smallest of the four rotation angles is the rotation angle for UX_π, then $\mathbf{v}$ is in $\mathbb{B}_x$ (red), and so forth.

The frames U, UX_π, UY_π, UZ_π all correspond to the same triple of principal axes (Fig. 3). The set $\mathbb{B}_w$ consists of the frames U for which U is the 'best' (i.e. closest to the identity frame, that is, smallest rotation angle) of the frames U, UX_π, UY_π, UZ_π. The set $\mathbb{B}_x$ consists of the frames U for which UX_π is the best of $U, UX_\pi, UY_\pi, UZ_\pi$. And so forth. Any one of the sets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ can be regarded as the set of all principal axis triples. Hence any one of them can be thought of as the space of double couple moment tensors.

Table 1. Some sets of frames, described both in frame coordinates ξ, ξ_x, ξ_y, ξ_z and in cartesian coordinates x, y, z. In the second column, $\mathbf{v} = (x, y, z)$ and $w = \sqrt{1 - \|\mathbf{v}\|^2}$.

ξ, ξ_x, ξ_y, ξ_z	x, y, z	in $\mathbb{B}$	Name
$\xi = t$	$\|\mathbf{v}\| = \sin(t/2)$	sphere	$\mathbb{S}(t)$
$\xi_x = t$	$\lvert x\rvert = \cos(t/2)$, $\|\mathbf{v}\| \leq 1$	two discs, Fig. 4	
$\xi_z \geq t$	$\lvert z\rvert \leq \cos(t/2)$, $\|\mathbf{v}\| \leq 1$	slab, Fig. 4	
$\min(\xi_x, \xi_y, \xi_z) = t$	$\max(\lvert x\rvert, \lvert y\rvert, \lvert z\rvert) = \cos(t/2)$, $\|\mathbf{v}\| \leq 1$	Fig. 9	$\mathbb{C}(t) \cap \mathbb{B}$
$\xi = \xi_x$	$2x^2 + y^2 + z^2 = 1$	ellipsoid, Fig. 5	
$\xi_0 = \xi$	$w \geq \max(\lvert x\rvert, \lvert y\rvert, \lvert z\rvert)$	Fig. 5	$\mathbb{B}_w$
$\xi_0 = \xi_x$	$\lvert x\rvert \geq \max(w, \lvert y\rvert, \lvert z\rvert)$	Fig. 5	$\mathbb{B}_x$
$\xi_0 = t$	$\max(w, \lvert x\rvert, \lvert y\rvert, \lvert z\rvert) = \cos(t/2)$	Fig. 10	$\mathbb{L}(t)$
$\xi_0 = \xi = t$	$\max(\lvert x\rvert, \lvert y\rvert, \lvert z\rvert) \leq w = \cos(t/2)$	Fig. 8	$\mathbb{L}_w(t)$

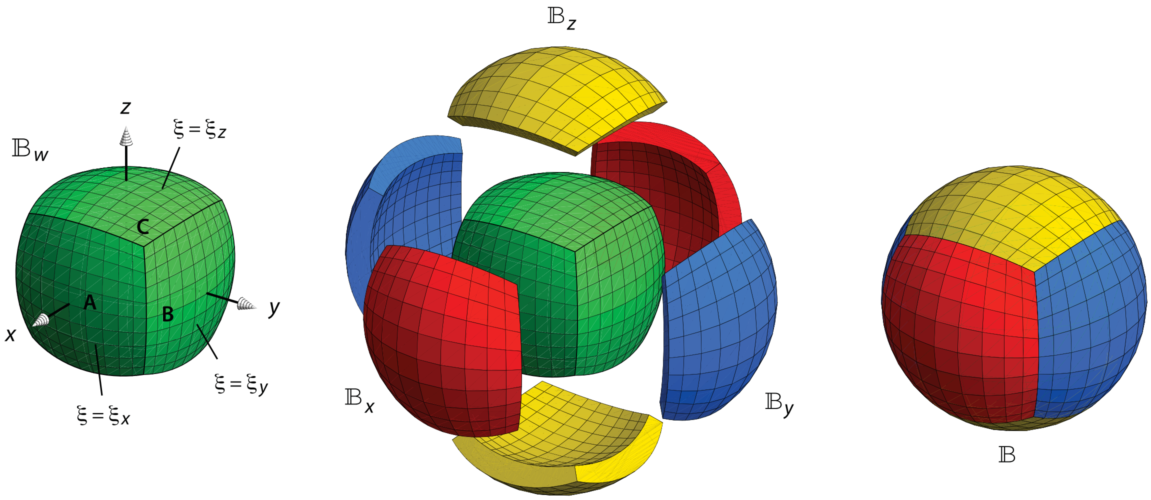

Figure 5. Partition of the unit ball $\mathbb{B}$ into subsets $\mathbb{B}_w$ (green), $\mathbb{B}_x$ (red), $\mathbb{B}_y$ (blue), and $\mathbb{B}_z$ (yellow). The sets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ consist of the frames U for which the principal axis angle ξ_0 is ξ, ξ_x, ξ_y, ξ_z, respectively. The set $\mathbb{B}_w$ is bounded by the ellipsoids $\xi = \xi_x$, $\xi = \xi_y$, and $\xi = \xi_z$. From eq. (43), the points **A**, **B**, **C** on the ellipsoid $\xi = \xi_x$ are $\mathbf{A} = (\frac{1}{\sqrt{2}}, 0, 0)$, $\mathbf{B} = (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, 0)$, and $\mathbf{C} = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Any one of $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ can be thought of as the set of principal axis triples, and hence as the set of double couple moment tensors. But $\mathbb{B}_w$ is special; for frames U in $\mathbb{B}_w$, the principal axis angle is the rotation angle of U itself.

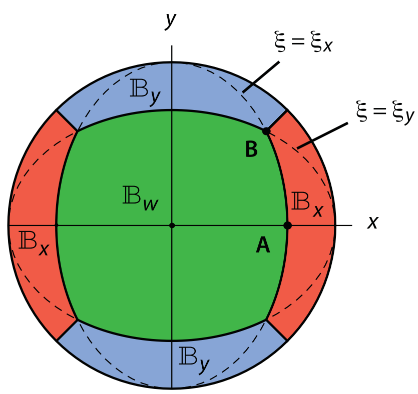

Figure 6. Plane section $z = 0$ of the unit ball $\mathbb{B}$ as partitioned in Fig. 5. The section of $\mathbb{B}_w$ consists of the points that are on or inside both ellipses $\xi = \xi_x$ and $\xi = \xi_y$ (eq. 43). This two dimensional diagram misses the complexity that occurs near the vertices of $\mathbb{B}_w$ (points like **C** in Fig. 5), where all four subsets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ meet.

Finally, using eqs (38) and (40), we can characterize the frames in $\mathbb{B}_w$ as those frames that are at least as close to I as to X_π, Y_π, or Z_π. The frames in $\mathbb{B}_x$ are those that are at least as close to X_π as to I, Y_π, Z_π. And so forth.

The above 'partition' of $\mathbb{B}$ is not quite a partition in the technical set-theoretic sense, since the pairwise intersections of the four sets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ are not empty. But they are nearly empty, being contained in the boundaries of the four sets. The partition can be turned into a true partition by carefully redefining the four sets on their boundaries, but we do not do so here. (For another partition, the redefinition is done in Appendix B of Tape & Tape (2012).)

3.9 Four symmetries of $\mathbb{U}$

Each of the symmetry mappings $U \to UX_\pi$, $U \to UY_\pi$, $U \to UZ_\pi$ permutes the subsets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ among each other. Eqs (37) and (41) show, for example, that the mapping $U \to UX_\pi$ swaps $\mathbb{B}_w$ with $\mathbb{B}_x$ and swaps $\mathbb{B}_y$ with $\mathbb{B}_z$. Since multiplication by any rotation preserves the natural (i.e. Haar) measure μ on the group of rotations, then each of $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ has the same measure. And since any two of the sets overlap only on their boundaries, which have measure zero, then the probability that a random rotation U lies in any one of $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ is the same for each, namely $\mu = 1/4$. And since each of the symmetry mappings preserves the metric d (eqs 20a), then the sets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ are all congruent with respect to d.

4 LEVEL SURFACES OF ξ_0

We let $\mathbb{L}(t)$ be the level surface of ξ_0 for the value t. That is,

$$\mathbb{L}(t) = \{\xi_0 = t\}. \tag{44}$$

In words, $\mathbb{L}(t)$ consists of all frames whose principal axis angle ξ_0 is t. When considered as subsets of the unit ball $\mathbb{B}$, the level surfaces make up all of $\mathbb{B}$:

$$\mathbb{B} = \bigcup_t \mathbb{L}(t). \tag{45}$$

Knowing all the level surfaces of ξ_0 is equivalent to knowing the function ξ_0 itself.

In principle the level surfaces of ξ_0 are simple. Since $\xi_0 = \min(\xi, \xi_x, \xi_y, \xi_z)$ (eq. 40), then in $\mathbb{B}_w$ the level surfaces of ξ_0 must coincide with the level surfaces of ξ, and in $\mathbb{B}_x$ they must coincide with the level surfaces of ξ_x, and so forth. The level surfaces of ξ are spheres centred at the origin, and the level surfaces of ξ_x, ξ_y, and ξ_z are planes perpendicular to the x-, y-, and z-axes, respectively (Section 3.7). The details can nevertheless stand some elaboration.

We define subsets $\mathbb{L}_w(t)$, $\mathbb{L}_x(t)$, $\mathbb{L}_y(t)$, $\mathbb{L}_z(t)$ of $\mathbb{L}(t)$ by

$$\begin{aligned} \mathbb{L}_w(t) &= \mathbb{B}_w \cap \mathbb{L}(t) = \{\xi_0 = \xi = t\}, \\ \mathbb{L}_x(t) &= \mathbb{B}_x \cap \mathbb{L}(t) = \{\xi_0 = \xi_x = t\}, \text{ etc.} \end{aligned} \tag{46}$$

In Section 3.8 we said that any one of $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ could be thought of as the set of all principal axis triples (or double couple moment tensors). Similarly, any one of $\mathbb{L}_w(t)$, $\mathbb{L}_x(t)$, $\mathbb{L}_y(t)$, $\mathbb{L}_z(t)$ can be thought of as the set of principal axis triples having principal axis angle $\xi_0 = t$.

4.1 In two dimensions

Fig. 7(a) shows how to construct the set $\mathbb{L}_w(t)$ in the $z = 0$ section of the unit ball. Figs 7(b) and (c) do the same for $\mathbb{L}_x(t)$ and $\mathbb{L}_y(t)$. For the t-value under consideration, namely $t = 100^\circ$, the level set $\mathbb{L}(t) = \mathbb{L}_w(t) \cup \mathbb{L}_x(t) \cup \mathbb{L}_y(t)$ consists of four little isosceles right 'triangles' having curved hypotenuses.

4.2 In three dimensions

The nature of $\mathbb{L}(t)$, whether in two or three dimensions, is determined by the relative disposition of a certain sphere $\mathbb{S}(t)$ and cube $\mathbb{C}(t)$. They, as well as the related solids $\widetilde{\mathbb{S}}(t)$ and $\widetilde{\mathbb{C}}(t)$, are defined in cartesian coordinates by

$$
\begin{aligned}
\mathbb{S}(t) &= \left\{ \|\mathbf{v}\| = \sin\frac{t}{2} \right\}, \\
\widetilde{\mathbb{S}}(t) &= \left\{ \|\mathbf{v}\| \geq \sin\frac{t}{2} \right\}, \\
\mathbb{C}(t) &= \left\{ \max(|x|, |y|, |z|) = \cos\frac{t}{2} \right\}, \\
\widetilde{\mathbb{C}}(t) &= \left\{ \max(|x|, |y|, |z|) \leq \cos\frac{t}{2} \right\}.
\end{aligned}
\tag{47}
$$

Thus the sphere $\mathbb{S}$(t) has radius $\sin(t/2)$, and the cube $\mathbb{C}$(t) has vertices $(\pm s, \pm s, \pm s)$, where $s = \cos(t/2)$. The set $\widetilde{\mathbb{S}}(t)$ consists of the points that are on or outside $\mathbb{S}(t)$, and $\widetilde{\mathbb{C}}(t)$ consists of the points that are on or inside $\mathbb{C}(t)$.

In frame coordinates, from eqs (12),

$$
\begin{aligned}
\mathbb{S}(t) &= \{\xi = t\}, \\
\widetilde{\mathbb{S}}(t) \cap \mathbb{B} &= \{\xi \geq t\}, \\
\mathbb{C}(t) \cap \mathbb{B} &= \{\min(\xi_x, \xi_y, \xi_z) = t\}, \\
\widetilde{\mathbb{C}}(t) \cap \mathbb{B} &= \{\min(\xi_x, \xi_y, \xi_z) \geq t\}.
\end{aligned}
\tag{48}
$$

In connection with $\mathbb{C}(t)$, we define points **D**, **E**, **F** by

$$
\begin{aligned}
\mathbf{D}(t) &= (s,\ 0,\ 0) \quad \text{(centre of a face of } \mathbb{C}(t)), \\
\mathbf{E}(t) &= (s,\ s,\ 0) \quad \text{(midpoint of an edge of } \mathbb{C}(t)), \\
\mathbf{F}(t) &= (s,\ s,\ s) \quad \text{(a vertex of } \mathbb{C}(t)), \\
s &= \cos\frac{t}{2}.
\end{aligned}
\tag{49}
$$

From eqs (46) and (48),

$$
\begin{aligned}
\mathbb{L}_w(t) &= \{\xi_0 = \xi = t\} \\
&= \{\xi = t\} \cap \{\min(\xi_x, \xi_y, \xi_z) \geq t\} \\
&= \mathbb{S}(t) \cap \widetilde{\mathbb{C}}(t) \cap \mathbb{B} \\
&= \mathbb{S}(t) \cap \widetilde{\mathbb{C}}(t).
\end{aligned}
\tag{50}
$$

Fig. 8 shows the evolution of $\mathbb{L}_w(t)$ with increasing t. According to eq. (50), the set $\mathbb{L}_w(t)$ is the portion of the sphere $\mathbb{S}(t)$ that is on or inside the cube $\mathbb{C}(t)$. As t increases, $\mathbb{S}(t)$ grows and $\mathbb{C}(t)$ shrinks. The set $\mathbb{L}_w(t)$ changes qualitatively when any of the points **D**, **E**, **F** is on $\mathbb{S}(t)$. From eqs (49),

$$
\begin{aligned}
\|\mathbf{D}(t)\| &= \sin(t/2) \text{ iff } t = t_1, \\
\|\mathbf{E}(t)\| &= \sin(t/2) \text{ iff } t = t_2, \\
\|\mathbf{F}(t)\| &= \sin(t/2) \text{ iff } t = t_3,
\end{aligned}
\tag{51}
$$

where

$$
\begin{aligned}
t_1 &= \pi/2, \\
t_2 &= \cos^{-1}(-1/3) = 109.5^\circ, \\
t_3 &= 2\pi/3.
\end{aligned}
\tag{52}
$$

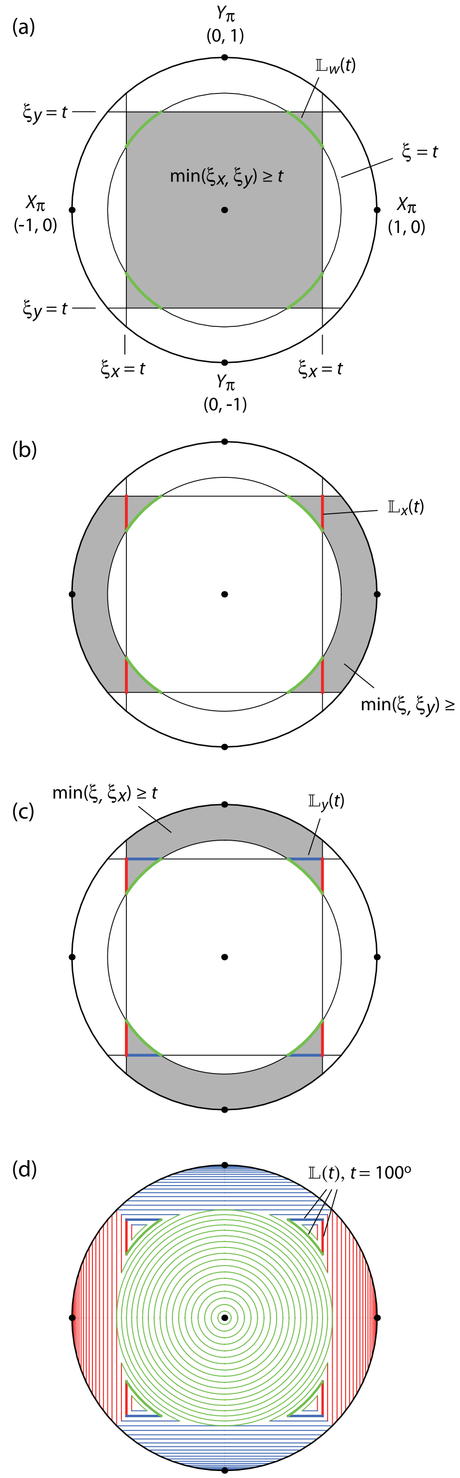

Figure 7. Constructing level curves $\mathbb{L}(t)$ of ξ_0 in the $z = 0$ section of the unit ball. (a) The set $\mathbb{L}_w(t)$ for $t = 100^\circ$. It consists of frames U such that $\xi_0 = \xi = t$, that is, $\xi = t \leq \min(\xi_x, \xi_y)$, where ξ, ξ_x, ξ_y are the rotation angles for U, UX_π, UY_π. Frames with $\xi_x \geq t$ make up a vertical strip, those with $\xi_y \geq t$ make up a horizontal strip, and the two strips intersect in the grey square, which is where $\min(\xi_x, \xi_y) \geq t$. Frames with $\xi = t$ make up the circle of radius $\sin(t/2)$. The intersection of the circle with the grey square is $\mathbb{L}_w(t)$ (green). (b) Similar but for $\mathbb{L}_x(t)$ (red), which consists of frames such that $\xi_0 = \xi_x = t$, that is, $\xi_x = t \leq \min(\xi, \xi_y)$. (c) For $\mathbb{L}_y(t)$ (blue). (d) Contour plot of ξ_0. Level curves $\mathbb{L}(t)$ are shown for $t = 0, 5, 10, \ldots 105^\circ$. The rotation angle ξ_z is equal to π everywhere on $z = 0$ and hence plays no role in these diagrams.

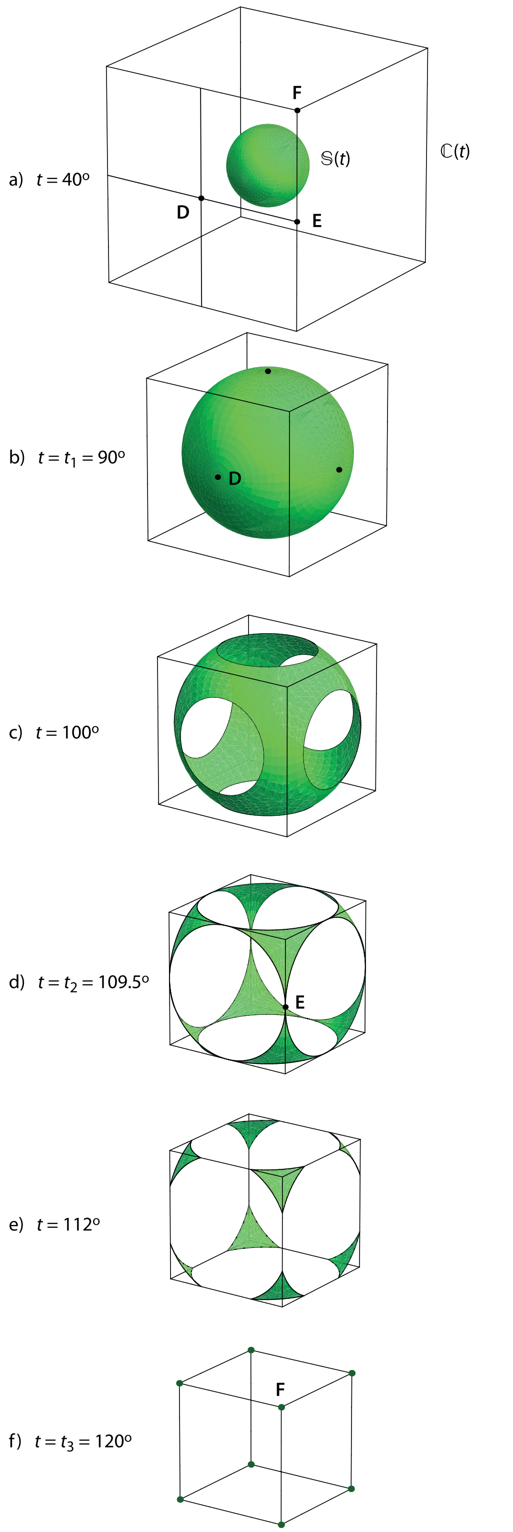

Figure 8. Sets $\mathbb{L}_w(t)$. For each t, the set $\mathbb{L}_w(t)$ consists of the frames U for which $\xi_0 = \xi = t$. It is the portion of the sphere $\mathbb{S}(t)$ that is on or inside the cube $\mathbb{C}(t)$ (eqs 47). As t increases, $\mathbb{S}(t)$ grows and $\mathbb{C}(t)$ shrinks. The nature of $\mathbb{L}_w(t)$ changes at $t = t_1$, when the face centres (e.g. $\mathbf{D}(t)$) of $\mathbb{C}(t)$ are on $\mathbb{S}(t)$. It changes again at $t = t_2$, when the edge midpoints (e.g. $\mathbf{E}(t)$) are on $\mathbb{S}(t)$. It changes once again at $t = t_3$, when the vertices (e.g. $\mathbf{F}(t)$) of $\mathbb{C}(t)$ are on $\mathbb{S}(t)$ and constitute all of $\mathbb{L}_w(t)$. The set $\mathbb{L}_w(t)$ is empty for larger t. The set $\mathbb{L}_w(t)$ can be regarded as the set of all principal axis triples with $\xi_0 = t$. The diagrams are all drawn at the same scale.

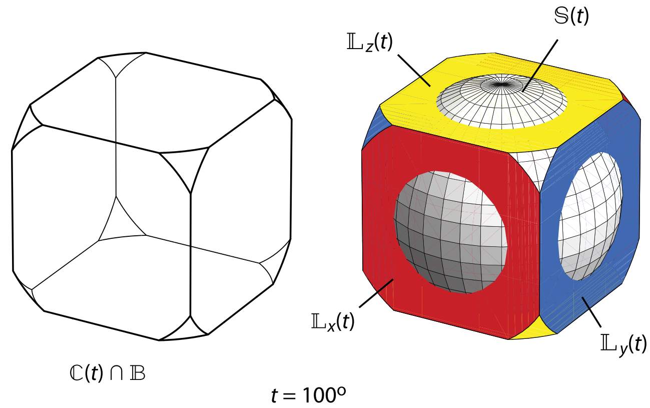

Figure 9. (*Left*) The intersection of the cube $\mathbb{C}(t)$ with the unit ball $\mathbb{B}$, for $t = 100°$. (*Right*) Sets $\mathbb{L}_x(t)$ (red), $\mathbb{L}_y(t)$ (blue), $\mathbb{L}_z(t)$ (yellow) for $t = 100°$. According to eq. (55), the union of these three sets is the portion of $\mathbb{C}(t) \cap \mathbb{B}$ that lies on or outside the sphere $\mathbb{S}(t)$.

The set $\mathbb{L}_w(t)$ is the sphere $\mathbb{S}(t)$ minus six open spherical caps. The caps are empty when $t \le t_1$, and they overlap when $t > t_2$. They completely cover the sphere when $t > t_3$, in which case $\mathbb{L}_w(t)$ is empty.

In particular, regardless of the direction of the rotation axis,

$$\xi \le \pi/2 \Rightarrow \xi_0 = \xi,$$
$$\xi > 2\pi/3 \Rightarrow \xi_0 < \xi. \tag{53}$$

(These also follow directly from eq. 34.) Also see Figs 13(a) and (d).

To describe $\mathbb{L}_x(t) \cup \mathbb{L}_y(t) \cup \mathbb{L}_z(t)$, we first find from eq. (46) that

$$\begin{aligned}\mathbb{L}_x(t) &= \{\xi_0 = \xi_x = t\} \\ &= \{\xi_x = t\} \cap \{\min(\xi, \xi_x, \xi_y, \xi_z) = t\} \\ &= \{\xi_x = t\} \cap \{\xi \ge t\} \cap \{\min(\xi_x, \xi_y, \xi_z) = t\}.\end{aligned} \tag{54}$$

Then, from eqs (48) and (54),

$$\begin{aligned}\mathbb{L}_x(t) \cup \mathbb{L}_y(t) \cup \mathbb{L}_z(t) &= \{\min(\xi_x, \xi_y, \xi_z) = t\} \cap \{\xi \ge t\} \\ &= \mathbb{C}(t) \cap \mathbb{B} \cap \widetilde{\mathbb{S}}(t).\end{aligned} \tag{55}$$

Thus $\mathbb{L}_x(t) \cup \mathbb{L}_y(t) \cup \mathbb{L}_z(t)$ is the portion of $\mathbb{C}(t) \cap \mathbb{B}$ that is on or outside $\mathbb{S}(t)$; see Fig. 9.

Eqs (50) and (55) together describe the level surface $\mathbb{L}(t)$:

$$\mathbb{L}(t) = \left(\mathbb{S}(t) \cap \widetilde{\mathbb{C}}(t)\right) \cup \left(\mathbb{C}(t) \cap \mathbb{B} \cap \widetilde{\mathbb{S}}(t)\right). \tag{56}$$

The evolution of the level surface $\mathbb{L}(t)$ with increasing t is illustrated in Fig. 10. Like $\mathbb{L}_w(t)$, the set $\mathbb{L}(t)$ changes qualitatively when $t = t_1, t_2$, or t_3.

Just as the symmetry mappings $U \to UX_\pi$, $U \to UY_\pi$, $U \to UZ_\pi$ permute the sets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ among each other (Section 3.9), so too they permute $\mathbb{L}_w(t)$, $\mathbb{L}_x(t)$, $\mathbb{L}_y(t)$, $\mathbb{L}_z(t)$. In spite of appearances, those latter four sets are therefore all congruent, but congruent with respect to the rotation distance d rather than with respect to the ordinary euclidean distance in $\mathbb{B}$. In Fig. 10(a), for example, since $t \le \pi/2$, each of $\mathbb{L}_w(t)$, $\mathbb{L}_x(t)$, $\mathbb{L}_y(t)$, $\mathbb{L}_z(t)$ is a d-sphere with radius equal to t; the sphere $\mathbb{L}_w(t)$ is centred at I, the sphere $\mathbb{L}_x(t)$ is centred at X_π, etc. In $\mathbb{B}$ the sphere $\mathbb{L}_x(t)$ appears to be two disconnected red discs, but in fact the discs are connected to each other along their boundaries, since antipodal points on the unit sphere are the same.

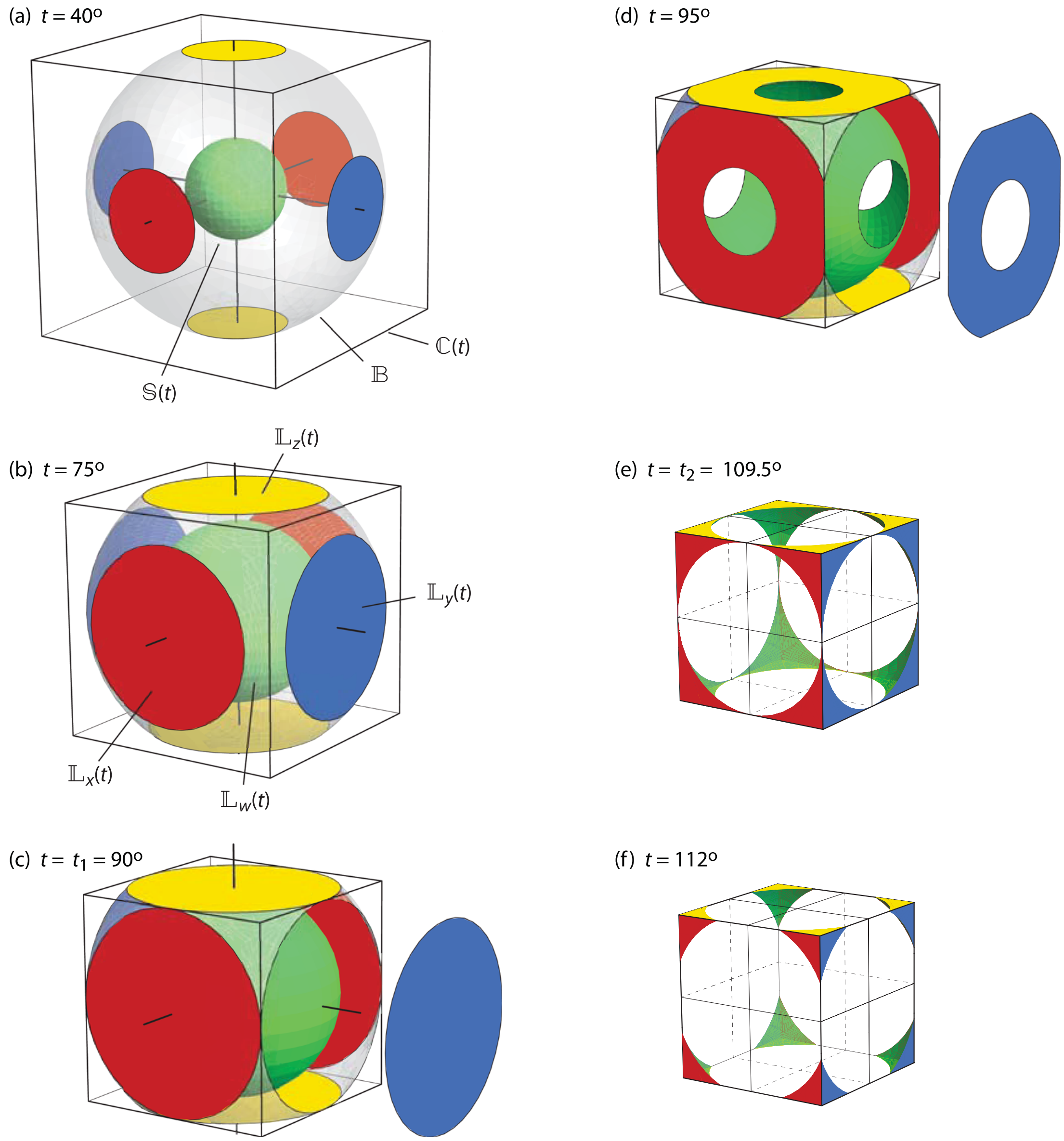

Figure 10. Level surfaces $\mathbb{L}(t)$ of ξ_0. Together, level surfaces like these constitute the contour plot of ξ_0, which is the three-dimensional analogue of Fig. 7(c). Each level set $\mathbb{L}(t)$ is the union of $\mathbb{L}_w(t)$ and $\mathbb{L}_x(t) \cup \mathbb{L}_y(t) \cup \mathbb{L}_z(t)$. The set $\mathbb{L}_w(t)$ is a subset of the sphere $\mathbb{S}(t)$ of radius $\sin(t/2)$, and $\mathbb{L}_x(t) \cup \mathbb{L}_y(t) \cup \mathbb{L}_z(t)$ is a subset of the cube $\mathbb{C}(t)$ with vertices $(\pm s, \pm s, \pm s)$, where $s = \cos(t/2)$; $\mathbb{L}_w(t)$ is the portion of $\mathbb{S}(t)$ that is on or inside $\mathbb{C}(t)$, and $\mathbb{L}_x(t) \cup \mathbb{L}_y(t) \cup \mathbb{L}_z(t)$ is the portion of $\mathbb{C}(t) \cap \mathbb{B}$ that is on or outside $\mathbb{S}(t)$. The subsets $\mathbb{L}_w(t)$ (green), $\mathbb{L}_x(t)$ (red), $\mathbb{L}_y(t)$ (blue), $\mathbb{L}_z(t)$ (yellow) are where $\mathbb{L}(t)$ meets $\mathbb{B}_w, \mathbb{B}_x, \mathbb{B}_y, \mathbb{B}_z$, respectively (Fig. 5). All diagrams are reproduced at the same scale. The unit ball $\mathbb{B}$ is only shown where it is inside the cube $\mathbb{C}$(t). In (c) and (d) one piece of $\mathbb{L}_y(t)$ has been detached to show $\mathbb{L}_w(t)$ better. Any one of $\mathbb{L}_w(t), \mathbb{L}_x(t), \mathbb{L}_y(t), \mathbb{L}_z(t)$ can be thought of as the set of principal axis triples having principal axis angle $\xi_0 = t$.

4.3 The transition values t_1, t_2, t_3

There are many equivalent ways to describe the 'times' t_1, t_2, t_3 when the level set $\mathbb{L}(t)$ changes qualitatively. As in eqs (51), t_1, t_2, t_3 are the times when the respective centres, edge midpoints, and vertices of the cube $\mathbb{C}(t)$ are on the sphere $\mathbb{S}(t)$. They are also the times when the points **A**, **B**, **C** of Fig. 5 are on $\mathbb{S}(t)$. The time $t = t_1$ is also when any two of the d-spheres $\xi = t, \xi_x = t, \xi_y = t, \xi_z = t$ are tangent to each other. The time $t = t_2$ is when any one of the d-spheres meets the intersection of two others tangentially. Note, however, that tangencies are not depicted correctly on the boundary of $\mathbb{B}$.

5 PROBABILITY OF ξ_0

In Appendix A we show that the probability density for ξ_0 is

$$p_{\xi_0}(t) = \frac{2}{\pi^2}\big(\text{area of } \mathbb{L}_w(t)\big). \tag{57}$$

That is, the probability that a random rotation have ξ_0 between 0 and t is

$$P(0 \le \xi_0 \le t) = \frac{2}{\pi^2}\int_0^t \big(\text{area of } \mathbb{L}_w(\xi)\big)\, d\xi. \tag{58}$$

The behaviour of $p_{\xi_0}(t)$ can therefore be seen at a glance in Fig. 8.

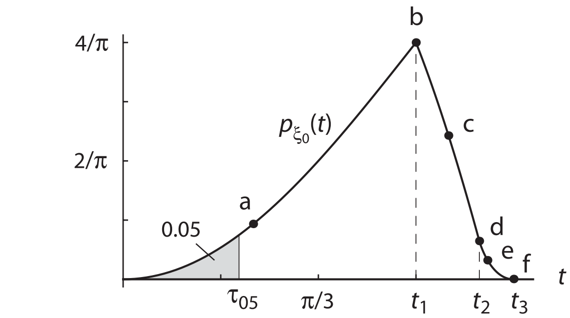

Figure 11. The probability density p_{ξ_0} for ξ_0. This curve shows what to expect for random orientations, and hence it serves as a reference against which distributions of empirical ξ_0 values can be compared. The density $p_{\xi_0}(t)$ is proportional to the area of $\mathbb{L}_w(t)$. The general shape of the curve p_{ξ_0} can therefore be understood by examining Fig. 8; the points (a)–(f) here refer to the subplots in that figure. The fifth percentile $\tau_{05} = 35.7^\circ$ is indicated on the curve.

An exact calculation of $p_{\xi_0}(t)$ falls into three cases. If $t \le t_1$, then $\mathbb{L}_w(t)$ is the entire sphere $\mathbb{S}(t)$ of radius $\sin(t/2)$, whose area is easily found. If $t_1 \le t \le t_2$, then $\mathbb{L}_w(t)$ is the sphere $\mathbb{S}(t)$ minus six non-intersecting spherical caps. The portion of a sphere $x^2 + y^2 + z^2 = R^2$ extending between $z = z_1$ and $z = z_2$ has area $2\pi R|z_2 - z_1|$, so the top (excised) spherical cap of $\mathbb{S}(t)$, which extends from $z = \cos(t/2)$ to $z = \sin(t/2)$, has area $2\pi \sin(t/2)(\sin(t/2) - \cos(t/2))$. Hence the area of $\mathbb{L}_w(t)$ is again known. For $t_2 \le t \le t_3$ the area of $\mathbb{L}_w(t)$ is found in Appendix B. Then, from eq. (57),

$$p_{\xi_0}(t) = \begin{cases} \dfrac{4}{\pi}(1 - \cos t), & 0 \le t \le t_1, \\[2ex] \dfrac{4}{\pi}(-2 + 2\cos t + 3\sin t), & t_1 \le t \le t_2, \\[2ex] \dfrac{48 \sin t}{\pi^2}\left(\theta - \tan\dfrac{t}{2}\,\sin^{-1}\left(\dfrac{\sin\theta}{\sqrt{2}}\right)\right)\Bigg|_{\theta=\theta_0}^{\theta=\pi/4}, & t_2 \le t \le t_3, \end{cases} \qquad (59)$$

where θ_0 is given in eq. (B4). Fig. 11 is a plot of $p_{\xi_0}(t)$ using eqs (59).

Eqs (59) agree with eqs (4) and (5) of Grimmer (1979a) and with eqs (3.1) of Kagan (1990). Eq. (57) is essentially a special case of a result stated by Grimmer (1979a).

The distinctive and highly non-uniform shape of the curve p_{ξ_0} should be borne in mind when interpreting distributions of empirical ξ_0 values. In Fig. 11 the fifth percentile τ_{05} is indicated on the curve. The figure shows that, for random rotations, one would expect only five per cent of them to have ξ_0 values less than $\tau_{05} = 35.6^\circ$. Thus any empirical distribution of ξ_0 that was even moderately skewed to the left would indicate much stronger orientation similarities than might be naively expected. This was clear to Kagan two decades ago (e.g. Kagan 1992), but the point bears repeating.

Hardebeck (2006) calculated ξ_0 values for moment tensors of many small earthquakes in California. She concluded that the moment tensor orientations for events close to each other in space were generally much more alike than those for events that were widely separated. To do so, she sorted event pairs into 17 subsets according to their distance from each other. For each subset, she then constructed a histogram of ξ_0 values. The histograms for distant events resemble the theoretical distribution for random rotations (our Fig. 11). The histograms for nearby events are skewed to the left, that is, skewed toward smaller ξ_0 values. See Hardebeck's fig. 3, especially the Southern California and Loma Prieta diagrams. Because small ξ_0 values are so unlikely for random orientations, the leftward skewed histograms are even more telling than they might first appear.

6 DEPICTING ROTATIONS IN RODRIGUES SPACE

In this paper we have depicted rotations as points in the unit ball $\mathbb{B}$. Some authors replace $\mathbb{B}$ with all of $\mathbb{R}^3$, which they then refer to as Rodrigues space (e.g. Frank 1988). Underlying each approach is a parameterization of rotations. The parameter points $\mathbf{v} = (x, y, z)$ and $\mathbf{v}' = (x', y', z')$ for a rotation U are

$$\begin{aligned} \mathbf{v} &= \left(\sin\frac{\xi}{2}\right)\mathbf{u} \qquad \text{(for unit ball)}, \\ \mathbf{v}' &= \left(\tan\frac{\xi}{2}\right)\mathbf{u} \qquad \text{(for Rodrigues)}, \end{aligned} \qquad (60)$$

where ξ and $\mathbf{u}$ are the rotation angle and normalized rotation axis of U. The points $\mathbf{v}$ and $\mathbf{v}'$ are then related by

$$\mathbf{v}' = \left(\sec\frac{\xi}{2}\right)\mathbf{v} = \frac{\mathbf{v}}{(1 - \|\mathbf{v}\|^2)^{1/2}}, \qquad (61a)$$

$$\mathbf{v} = \left(\cos\frac{\xi}{2}\right)\mathbf{v}' = \frac{\mathbf{v}'}{(1 + \|\mathbf{v}'\|^2)^{1/2}}. \qquad (61b)$$

From eq. (61a) the Rodrigues coordinates x', y', z' are related to our usual x, y, z coordinates by

$$x' = x/w, \quad y' = y/w, \quad z' = z/w. \qquad (62)$$

The Rodrigues setting has some advantages. In Rodrigues space, with distances between rotations measured by the rotation metric d (eq. 18), the locus of rotations equidistant from two given rotations turns out to consist of two planes. In Rodrigues space the set $\mathbb{B}_w$, whose boundary surfaces are formed by $\xi = \xi_x$, $\xi = \xi_y$, and $\xi = \xi_z$, must therefore be a solid polyhedron, with truly planar faces. In fact it is the solid unit cube; the boundary surface $\xi = \xi_x$, for example, is $|x'| = 1$, since

$$|x'| = \frac{|x|}{w} = \frac{\cos(\xi_x/2)}{\cos(\xi/2)}. \qquad (63)$$

But Rodrigues space also has its drawbacks. It does not contain 180° rotations, and such rotations therefore have to be treated as limiting cases. And although some sets, like $\mathbb{B}_w$, look better in Rodrigues space than in the unit ball, other sets look worse. The level surfaces $\xi_x = t$, $\xi_y = t$, and $\xi_z = t$, which were planar in the unit ball (e.g. Fig. 4), become hyperboloids in Rodrigues space. Thus the sets $\mathbb{L}_x(t)$, $\mathbb{L}_y(t)$, $\mathbb{L}_z(t)$, which were parts of cube faces in Fig. 10, look exotic in Rodrigues space (Fig. 12). And eq. (57) is an example of a result that is better expressed in the context of the unit ball than in Rodrigues space. In any event, one can easily go back and forth between the (interior of the) unit ball and Rodrigues space, using eqs (61).

Frank (1988) has a readable exposition of Rodrigues space, and details can be found in Morawiec (2004). In Frank's terminology, our set $\mathbb{B}_w$ is the fundamental zone for the group $\mathbb{D}_2$. Our angle ξ_0 is the misorientation angle for $\mathbb{D}_2$.

7 THE ANGLE ξ_0 FOR FIXED ξ

For purposes of comparing ξ_0 with other measures of orientation separation, it is convenient to plot contours of ξ_0 on the spheres $\mathbb{S}(t)$. For each t, $0 \le t \le \pi$, the contour plot on $\mathbb{S}(t)$ displays ξ_0 as a function of rotation axis, with rotation angle $\xi = t$ fixed. Due to the

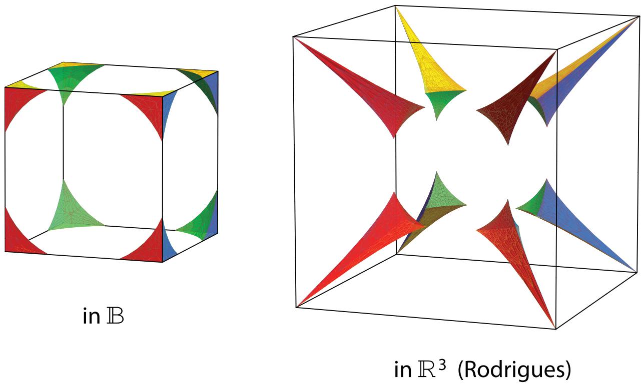

Figure 12. Level surface $\mathbb{L}(t)$ of ξ_0 for $t = 112°$, depicted both in the unit ball $\mathbb{B}$ and in Rodrigues space $\mathbb{R}^3$. The subset $\mathbb{L}_w(t)$ (green), which is part of a sphere when depicted in $\mathbb{B}$, is part of a larger sphere in Rodrigues space. (The diagrams are drawn at different scales.) The subset $\mathbb{L}_x(t)$ (red), which is a subset of opposite cube faces in $\mathbb{B}$, is a subset of a hyperboloid in Rodrigues space. Similarly for $\mathbb{L}_y(t)$ (blue) and $\mathbb{L}_z(t)$ (yellow). The peculiar cast to some of the colours is due to the neutral lighting that was used to help with the perspective; the only colours present are green, red, blue, and yellow. For the set shown here, depiction in the unit ball is superior to depiction in Rodrigues space, but for other sets the reverse can be true.

absolute values in the characterization of ξ_0 (eq. 34), it is enough to depict ξ_0 in one octant of $\mathbb{B}$, as in Fig. 13.

8 THE ANGLE ξ_0 AS A FUNCTION OF MOMENT TENSORS

For moment tensors M_1 and M_2 having distinct eigenvalues, the angle ξ_0 between them is defined by choosing frames U_1 and U_2 associated with the T, B, and P directions of the moment tensors and then calculating ξ_0 between U_1 and U_2. Formally:

$$\xi_0([\mathbf{\Lambda}_1]_{U_1}, [\mathbf{\Lambda}_2]_{U_2}) = \xi_0(U_1, U_2), \quad \mathbf{\Lambda}_1, \mathbf{\Lambda}_2 \in \mathbb{W}_0, \tag{64}$$

where $\xi_0(U_1, U_2)$ is given by eq. (34) and where, with $\mathbf{\Lambda} = (\lambda_1, \lambda_2, \lambda_3)$,

$$\mathbb{W}_0 = \{\mathbf{\Lambda} : \lambda_1 > \lambda_2 > \lambda_3\},$$
$$[\mathbf{\Lambda}] = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix},$$
$$[\mathbf{\Lambda}]_U = U[\mathbf{\Lambda}]U^{-1}. \tag{65}$$

Any moment tensor M with distinct eigenvalues can be written in the form $M = [\mathbf{\Lambda}]_U$, with $\mathbf{\Lambda} \in \mathbb{W}_0$. The expression is unique except that $[\mathbf{\Lambda}]_U = [\mathbf{\Lambda}]_{UR}$ for $R \in \mathbb{D}_2 = \{I, X_\pi, Y_\pi, Z_\pi\}$. Thus in eq. (64) there are four choices for each of U_1 and U_2, but the outcome $\xi_0(U_1, U_2)$ is the same for all, due to eq. (30).

We next give an example suggesting that ξ_0 may not always be the most reasonable measure of orientation difference for moment tensors. Fig. 14 shows double couple moment tensors $M_1 = [\mathbf{\Lambda}]_{U_1}$ and $M_2 = [\mathbf{\Lambda}]_{U_2}$ at their natural locations in the unit ball. The reference double couple $M_0 = [\mathbf{\Lambda}]$ is at the origin. The orientations U_1 and U_2 are 90° rotations, and hence $\xi_0(I, U_1) = \xi_0(I, U_2) = \pi/2$ (eq. 53). Then $\xi_0(M_0, M_1) = \xi_0(M_0, M_2) = \pi/2$, by eq. (64). But the moment tensor M_1 looks not so different from M_0, especially in the x-direction, where both are red, whereas M_2 and M_0 look as different as they can be and still be the same size. Since all three moment tensors have the same eigenvalue triple $\mathbf{\Lambda} = (1, 0, -1)$, the differences in appearance ought to be due to differences in orientation. Thus ξ_0 is not capturing our informal sense of difference in moment tensor orientations. Willemann (1993) makes the same point when, referring to ξ_0, he says

> . . . the rotation required for alignment is unreasonable in the sense that a rotation of only 90° is required to align opposite double couples.

8.1 An alternative to ξ_0

In trying to define a reasonable measure of difference in moment tensor orientations, we are motivated by the fact that the eigenvalues of a moment tensor give its size and source type (its pattern, in beachball language), and that the eigenvectors give the orientation. For measuring differences in orientation, the eigenvalues should therefore be ignored (but see below). In the definition of ξ_0 for moment tensors (eq. 64), the only role of the eigenvalues is to order the principal axes, so that the principal axes of the one moment tensor can be associated with the principal axes of the other. Once the association is made, it does not matter which of the three axes is the T, B, or P axis.

That may be going too far, as was suggested in Fig. 14. The parameter ω, defined next, gives an alternative measure of orientation differences that does capture some of the intuition for the differences seen in Fig. 14. For orientations (i.e. frames) U_1 and U_2, the function ω is defined by

$$\omega(U_1, U_2) = \angle([\mathbf{\Lambda}_0]_{U_1}, [\mathbf{\Lambda}_0]_{U_2}), \quad \mathbf{\Lambda}_0 = (1, 0, -1), \tag{66}$$

where $\angle$ is the angle in the space of 3×3 matrices. That is, 3×3 matrices are treated as elements of $\mathbb{R}^9$, and the angle between them is computed using the standard inner product in $\mathbb{R}^9$. Thus, for 3×3 matrices M and N,

$$\cos \angle(M, N) = \frac{M \cdot N}{\|M\|\|N\|} = \frac{\sum m_{ij}\, n_{ij}}{\left(\sum m_{ij}^2\right)^{1/2} \left(\sum n_{ij}^2\right)^{1/2}}. \tag{67}$$

As was done earlier for ξ_0, we write $\omega(U)$ for $\omega(I, U)$.

For the 90° rotations U_1 and U_2 in Fig. 14, we find from eq. (66) that $\omega(U_1) = \pi/3$ and $\omega(U_2) = \pi$, which is more in keeping with our informal sense of the orientation differences than the values $\xi_0(U_1) = \xi_0(U_2) = \pi/2$. More generally, Fig. 15 shows how ξ_0 and ω compare for 90° rotations.

Since $[\mathbf{\Lambda}]_{UR} = [\mathbf{\Lambda}]_U$ for $R \in \mathbb{D}_2$, then

$$\omega(U_1 R_1, U_2 R_2) = \omega(U_1, U_2), \qquad R_1, R_2 \in \mathbb{D}_2. \tag{68}$$

This means that ω can be defined for moment tensors, not just for frames. The definition is analogous to eq. (64):

$$\omega([\mathbf{\Lambda}_1]_{U_1}, [\mathbf{\Lambda}_2]_{U_2}) = \omega(U_1, U_2), \quad \mathbf{\Lambda}_1, \mathbf{\Lambda}_2 \in \mathbb{W}_0, \tag{69}$$

or, equivalently,

$$\omega([\mathbf{\Lambda}_1]_{U_1}, [\mathbf{\Lambda}_2]_{U_2}) = \angle([\mathbf{\Lambda}_0]_{U_1}, [\mathbf{\Lambda}_0]_{U_2}), \quad \mathbf{\Lambda}_1, \mathbf{\Lambda}_2 \in \mathbb{W}_0. \tag{70}$$

For any $\mathbf{\Lambda} \in \mathbb{W}_0$ the closest double couple to the moment tensor $[\mathbf{\Lambda}]_U$ is $[\mathbf{\Lambda}_{\mathrm{DC}}]_U$, where $\mathbf{\Lambda}_{\mathrm{DC}}$ is the vector projection of $\mathbf{\Lambda}$ onto $\mathbf{\Lambda}_0$. That is,

$$\mathbf{\Lambda}_{\mathrm{DC}} = \frac{\mathbf{\Lambda} \cdot \mathbf{\Lambda}_0}{\|\mathbf{\Lambda}_0\|^2}\, \mathbf{\Lambda}_0. \tag{71}$$

Since $\mathbf{\Lambda}_{\mathrm{DC}}$ and $\mathbf{\Lambda}_0$ are scalar multiples of each other, then eq. (70) can be rewritten as

$$\omega([\mathbf{\Lambda}_1]_{U_1}, [\mathbf{\Lambda}_2]_{U_2}) = \angle([\mathbf{\Lambda}_{\mathrm{DC}}]_{U_1}, [\mathbf{\Lambda}_{\mathrm{DC}}]_{U_2}), \quad \mathbf{\Lambda}_1, \mathbf{\Lambda}_2 \in \mathbb{W}_0. \tag{72}$$

In words, $\omega(M_1, M_2)$ is the angle, in matrix space, between the closest double couples to M_1 and M_2. The intuition behind ω is that,

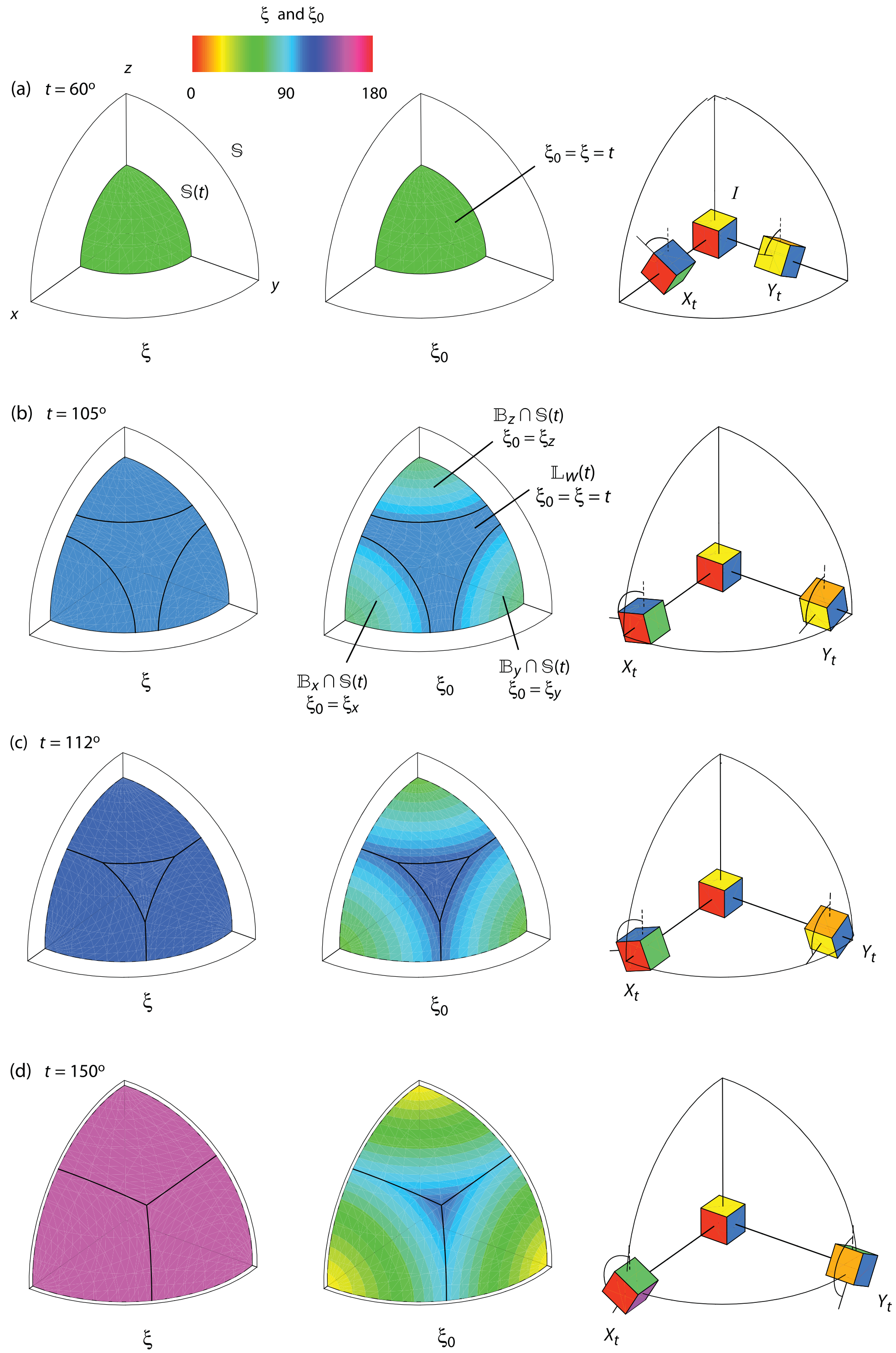

Figure 13. Contour plots of the rotation angle ξ and the principal axis angle ξ_0 on one octant of the sphere $\mathbb{S}(t)$. On $\mathbb{S}(t)$ the angle ξ coincides with t, and the contour plot for ξ is a solid colour in each case (left column). On the subset $\mathbb{L}_w(t)$ the angle ξ_0 coincides with ξ, and so the contour plot for ξ_0 (middle column) on $\mathbb{L}_w(t)$ is a solid colour matching that for ξ. On $\mathbb{B}_x \cap \mathbb{S}(t)$ the angle ξ_0 coincides with ξ_x, whose contour surfaces in $\mathbb{B}$ are planes perpendicular to the x-axis and whose contour curves on $\mathbb{S}(t)$ are therefore circles with poles on the x-axis. Similarly on $\mathbb{B}_y \cap \mathbb{S}(t)$ and $\mathbb{B}_z \cap \mathbb{S}(t)$. The set $\mathbb{L}_w(t)$ varies with t as shown in Fig. 8 and as described in eq. (50). The right-hand diagrams show the rotations X_t and Y_t at their correct locations in $\mathbb{B}$.

first, by using angles (in matrix space) rather than distances, we ignore all information about size. Second, by replacing $\mathbf{\Lambda}$ with $\mathbf{\Lambda}_0$ (or $\mathbf{\Lambda}_{\mathrm{DC}}$), we ignore the information about source type. The remaining information should be orientation. However, the distinction between the T, B, and P axes is not lost, due to the presence of $\mathbf{\Lambda}_0$ in eq. (70), and the three axes get treated differently in calculating ω. In this regard, ω differs from ξ_0.

It is clear from eqs (64) and (69) that $\xi_0([\mathbf{\Lambda}_1]_{U_1}, [\mathbf{\Lambda}_2]_{U_2})$ and $\omega([\mathbf{\Lambda}_1]_{U_1}, [\mathbf{\Lambda}_2]_{U_2})$ are both independent of the eigenvalue triples $\mathbf{\Lambda}_i \in \mathbb{W}_0$. That fact can be disconcerting at first, since changing

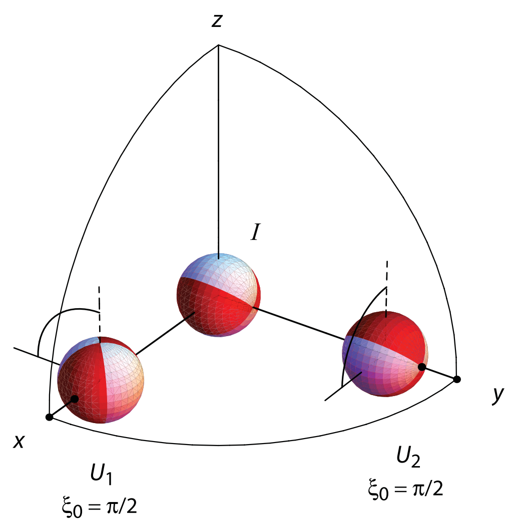

Figure 14. Double couple beachballs (moment tensors) $M_1 = [\mathbf{\Lambda}_0]_{U_1}$ and $M_2 = [\mathbf{\Lambda}_0]_{U_2}$, where $\mathbf{\Lambda}_0 = (1, 0, -1)$ and where U_1 and U_2 are 90° rotations about the x- and y-axes, respectively. The moment tensor $M_0 = [\mathbf{\Lambda}_0]_I = [\mathbf{\Lambda}_0]$ is at the origin. The measure ξ_0 considers the difference in orientation between M_0 and M_1 to be the same as that between M_0 and M_2, namely $\xi_0 = \pi/2$ in both cases. Yet, intuitively, the beachball M_1 resembles M_0 more closely than does M_2. In fact, since their colours are reversed, M_2 and M_0 are as different as moment tensors of the same size can be, whereas M_0 and M_1 are not so different from each other, especially in the x direction, where both are red. Since all three beachballs have the same eigenvalue triple $\mathbf{\Lambda}_0$, any differences in appearance ought to be attributable to differences in orientation. Since the parameter ξ_0 does not capture these differences, ξ_0 may not always be the best measure of separation for moment tensor orientations.

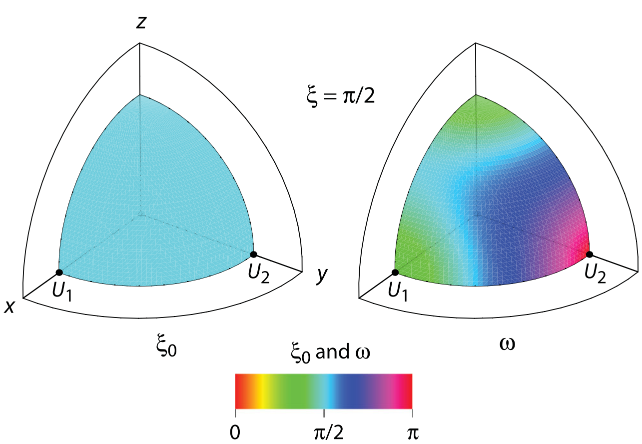

Figure 15. Comparison of ξ_0 and ω for 90° rotations. The angle ξ_0 is constant, $\xi_0 = \pi/2$, whereas ω varies from $\pi/3$ at U_1 to π at U_2. The orientations U_1 and U_2 are the 90° rotations about the x- and y-axes, the same as in Fig. 14. The ω values for U_1 and U_2 seem to capture our intuition from Fig. 14 better than do the ξ_0 values.

the $\mathbf{\Lambda}_i$ can drastically change the size and colouring patterns of the moment tensor beachballs. But both ξ_0 and ω, after all, were intended to be measures of differences in moment tensor orientation, nothing more.

Several authors have used the angle $\angle(M, N)$ in matrix space (eq. 67) as a measure of separation between moment tensors M and N (e.g. Willemann 1993; Frohlich & Davis 1999; Hjörleifsdóttir & Ekström 2010). That measure incorporates moment tensor source type as well as orientation (but not size). Our parameter ω is based on ideas of Willemann (1993).

In a separate paper we will study ω in detail and further compare it with ξ_0.

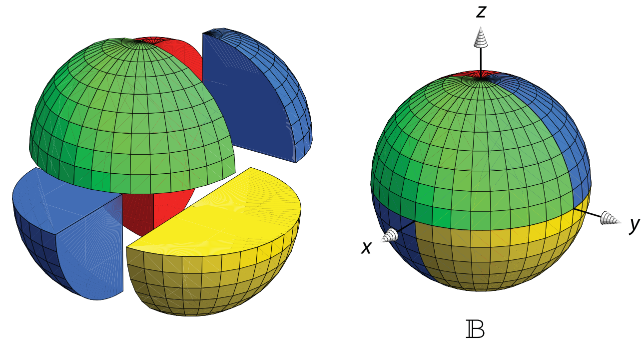

Figure 16. A partition of the unit ball $\mathbb{B}$ into four subsets. Three of them are quadrants (green, red, yellow), and the fourth is the union of two opposite octants (blue). Eqs (11) show that the symmetry mappings $U \to UX_\pi$, $U \to UY_\pi$, $U \to UZ_\pi$ permute the four subsets among each other. Just as in Fig. 5, any one of the four subsets can therefore be thought of as the set of principal axis triples and hence as the set of double couple moment tensors. What distinguishes the partition $\mathbb{B}_w, \mathbb{B}_x, \mathbb{B}_y, \mathbb{B}_z$ in Fig. 5 is its close connection to the function ξ_0.

9 OTHER PARTITIONS OF FRAMES

The partition $\mathbb{B}_w, \mathbb{B}_x, \mathbb{B}_y, \mathbb{B}_z$ of the unit ball $\mathbb{B}$ in Fig. 5 was constructed so that any one of the four subsets could represent the set of principal axis triples or, equivalently, the set of double couple moment tensors. Many other partitions can do the same.

Fig. 16 shows a partition of $\mathbb{B}$ into unnamed green, red, blue, and yellow subsets. Eq. (11b) shows that the mapping $U \to UX_\pi$ swaps the green with the red subset (and blue with yellow). Similarly, $U \to UY_\pi$ swaps green with blue, and $U \to UZ_\pi$ swaps green with yellow. Those properties by themselves are enough to ensure that the green subset can serve to represent all principal axis triples, since, for any frame U, one of U, UX_π, UY_π, UZ_π will be in the green subset. So the two partitions have much in common. Many other such partitions of rotations are also possible; Tape & Tape (2012) have one that is based on slip and dip angles. What distinguishes the partition $\mathbb{B}_w, \mathbb{B}_x, \mathbb{B}_y, \mathbb{B}_z$ is the defining condition $\xi_0 = \xi$ for $\mathbb{B}_w$.

10 OVERVIEW

The principal axis angle ξ_0 measures differences in moment tensor orientations. Conceptually, however, ξ_0 can stand alone as a measure of differences between principal axis triples, by which we mean ordered triples of mutually perpendicular lines, not necessarily associated with any moment tensor. To elucidate the logic of ξ_0, we have left moment tensors out of the discussion until the very end.

Informally, the angle ξ_0 between two principal axis triples is the smallest angle required to rotate one triple into the other. Formally, $\xi_0(U_1, U_2)$ is defined in Section 3.2 as a function of frames U_1 and U_2. The general case $\xi_0(U_1, U_2)$ is easily reduced to the case $\xi_0(I, U)$ where the first frame is the identity. Eq. (34) gives a concise formula for ξ_0.

The behaviour of ξ_0 is not at all transparent, especially if one relies on calculations alone. In Section 2.3 we describe how rotation matrices, or frames, can be depicted as points in the unit ball $\mathbb{B}$. We do not always distinguish between a point $\mathbf{v}$ of $\mathbb{B}$ and its corresponding rotation matrix U, and we do not always distinguish between $\mathbb{B}$ and the group $\mathbb{U}$ of rotations. The angle $\xi_0 = \xi_0(I, U)$ then becomes a function on $\mathbb{B}$ and can be visualized. Fig. 13 shows how ξ_0 varies for frames U having fixed rotation angle. Fig. 10 exhibits the level surfaces of ξ_0 in $\mathbb{B}$. It shows, for example, that it is easy to construct U with $\xi_0 = 40^\circ$ but difficult to construct U with $\xi_0 = 112^\circ$; in the latter case, both the rotation axis and the rotation angle of U must be chosen judiciously.

Each triple of principal axes corresponds to four frames (Fig. 3). The group $\mathbb{U}$ of frames is therefore too big by a factor of 4 for representing principal axis triples efficiently, or for representing double couple moment tensors. There are many ways to partition $\mathbb{U}$ into four subsets such that any one of the subsets represents all principal axis triples. One such partition, stated for the unit ball $\mathbb{B}$ rather than for $\mathbb{U}$, consists of the subsets $\mathbb{B}_w$, $\mathbb{B}_x$, $\mathbb{B}_y$, $\mathbb{B}_z$ shown in Fig. 5. Another such partition is shown in Fig. 16. What distinguishes the first partition is its close connection with the function ξ_0.

The set $\mathbb{L}_w(t)$ consists of the rotations whose principal axis angle ξ_0 and rotation angle ξ both equal t. When those rotations are considered as points in the unit ball, the set $\mathbb{L}_w(t)$ is a subset of the sphere of radius $\sin(t/2)$ (Fig. 8). In Appendix A we show that for random orientations the probability density $p_{\xi_0}(t)$ for ξ_0 is proportional to the area of $\mathbb{L}_w(t)$. We then calculate the area of $\mathbb{L}_w(t)$, which gives an analytical expression for p_{ξ_0}.

The curve p_{ξ_0} shows what to expect for random orientations (Fig. 11). It serves as a reference curve against which to compare distributions of empirical ξ_0 values. It shows that, for inferring similarities of orientations, small values of ξ_0 are more significant than they might appear.

The mathematical setting for discussing ξ_0 is the group $\mathbb{U}$ of frames. The distance d between two frames is defined to be the rotation angle of the rotation that takes the one frame to the other. Then d is indeed a metric for frames, and ξ_0 is the metric for principal axes triples that is the counterpart of d.

Fig. 14 shows that the angle ξ_0 sometimes conflicts with our intuitive feeling as to what constitutes a reasonable measure of separation for moment tensor orientations. We introduce an alternate measure ω and compare it with ξ_0. The angle ω between moment tensors M_1 and M_2 is the angle, in matrix space, between the closest double couples to M_1 and M_2.

ACKNOWLEDGMENTS

We thank two reviewers for helpful suggestions. One reviewer was Yan Kagan, and the other was anonymous. Both Hans Grimmer and Adam Morawiec kindly responded to our e-mail inquiries regarding the historical evolution of concepts related to the misorientation angle.

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APPENDIX A: PROBABILITY DENSITY $P_{\xi_0}(t)$ IS PROPORTIONAL TO THE AREA OF $\mathbb{L}_w(t)$

In this appendix we verify eq. (57) of Section 5.

Since the sets $\mathbb{L}_w(t)$, $\mathbb{L}_x(t)$, $\mathbb{L}_y(t)$, $\mathbb{L}_z(t)$ are permuted among each other by the symmetry mappings $U \to UX_\pi$, $U \to UY_\pi$, $U \to UZ_\pi$, so too are the four sets $\widehat{\mathbb{L}}_w(t)$, $\widehat{\mathbb{L}}_x(t)$, $\widehat{\mathbb{L}}_y(t)$, $\widehat{\mathbb{L}}_z(t)$ defined by

$$\begin{aligned} \widehat{\mathbb{L}}_w(t) &= \{\xi_0 = \xi \le t\} = \textstyle\bigcup_{t' \le t} \mathbb{L}_w(t'), \\ \widehat{\mathbb{L}}_x(t) &= \{\xi_0 = \xi_x \le t\} = \textstyle\bigcup_{t' \le t} \mathbb{L}_x(t'), \text{ etc.} \end{aligned} \tag{A1}$$

The four sets have the same (Haar) measure μ, since the symmetry mappings preserve the measure. The pairwise intersections of the sets have measure zero, and so the probability that a random rotation has $\xi_0 \le t$ is

$$P\big(\xi_0 \le t\big) = 4\mu\big(\widehat{\mathbb{L}}_w(t)\big), \tag{A2}$$

where μ is assumed to be normalized so that $\mu(\mathbb{B}) = 1$.

Since $\mathbb{L}_w(t') \subset \mathbb{S}(t')$ (e.g. eq. 50), then

$$\widehat{\mathbb{L}}_w(t) \cap \mathbb{S}(t') = \begin{cases} \mathbb{L}_w(t'), & t' \le t, \\ \emptyset, & t' > t. \end{cases} \tag{A3}$$

The probability that a random rotation has its rotation angle between ξ and $\xi + d\xi$ and has the longitude and colatitude of its rotation axis between ϕ and $\phi + d\phi$ and between θ and $\theta + d\theta$ is (e.g. Miles 1965)

$$\frac{1}{2\pi^2} \sin^2\frac{\xi}{2} \sin\theta \, d\theta \, d\phi \, d\xi. \tag{A4}$$

If F_ξ is the usual parameterization of the sphere $\mathbb{S}(\xi)$ of radius $\sin(\xi/2)$, that is, if

$$\begin{aligned} &F_\xi(\theta, \phi) = \sin\frac{\xi}{2}\,(\cos\phi\,\sin\theta,\ \sin\phi\,\sin\theta,\ \cos\theta), \\ &0 \le \theta \le \pi, \ \ 0 \le \phi \le 2\pi, \end{aligned} \tag{A5}$$

then the parameterization of the unit ball in terms of (ξ, θ, ϕ) is

$$(\xi, \theta, \phi) \to F_\xi(\theta, \phi). \tag{A6}$$

From eq. (A4), the measure of a set $\mathbb{A} \subset \mathbb{B}$ is

$$\begin{aligned} \mu(\mathbb{A}) &= \frac{1}{2\pi^2} \iiint\limits_{F_\xi(\theta,\phi) \in \mathbb{A}} \sin^2\frac{\xi}{2} \sin\theta \, d\theta \, d\phi \, d\xi \\ &= \frac{1}{2\pi^2} \int_{\xi=0}^{\xi=\pi} \iint\limits_{(\theta,\phi) \in F_\xi^{-1}(\mathbb{A} \cap \mathbb{S}(\xi))} \sin^2\frac{\xi}{2} \sin\theta \, d\theta \, d\phi \ d\xi \\ &= \frac{1}{2\pi^2} \int_0^\pi \text{area of } \big(\mathbb{A} \cap \mathbb{S}(\xi)\big)\, d\xi. \end{aligned} \tag{A7}$$

If we take $\mathbb{A} = \widehat{\mathbb{L}}_w(t)$ in eq. (A7), then eqs (A2) and (A3) give

$$P\big(\xi_0 \le t\big) = \frac{2}{\pi^2} \int_0^t \big(\text{area of } \mathbb{L}_w(\xi)\big)\, d\xi. \tag{A8}$$

The derivative of eq. (A8) gives the probability density for ξ_0:

$$p_{\xi_0}(t) = \frac{2}{\pi^2} \big(\text{area of } \mathbb{L}_w(t)\big), \tag{A9}$$

which is eq. (57).

To get a sense of how the invariant measure μ differs from the ordinary volume in $\mathbb{B}$, we make the substitution $\rho = \sin(\xi/2)$ in eq. (A7) and find

$$\mu(\mathbb{A}) = \frac{1}{\pi^2} \int_0^1 \frac{\text{area of } \big(\mathbb{A} \cap \mathbb{S}_e(\rho)\big)}{\sqrt{1 - \rho^2}}\, d\rho, \tag{A10}$$

where $\mathbb{S}_e(\rho)$ is the sphere of euclidean radius ρ. Without the factor $(1 - \rho^2)^{-1/2}$, the integral in eq. (A10) would give the ordinary volume of $\mathbb{A}$.

This is the natural place to calculate the distribution of the 'misorientation axes'—the rotation axes of the orientations in $\mathbb{B}_w$. We instead refer to Morawiec (1996).

APPENDIX B: AREA OF $\mathbb{L}_w(t)$ WHEN $t \ge t_2$

In this appendix we verify eq. (59) of Section 5.

The set H' shown in Fig. B1 is $1/48^{\text{th}}$ of the set $\mathbb{L}_w(t)$. It is the radial projection of the red planar set H to the sphere $S(t)$ of radius $\sin(t/2)$. The set H is on the cube face $x = \cos(t/2)$, and the radial projection $f\colon H \to H'$ is

$$f(y, z) = \left(\sin\frac{t}{2}\right) \frac{(\cos(t/2),\ y,\ z)}{\|(\cos(t/2),\ y,\ z)\|}. \tag{B1}$$

Then

$$\text{Area of H}' = \iint\limits_H \left\| \frac{\partial f}{\partial y} \times \frac{\partial f}{\partial z} \right\| dy\, dz. \tag{B2}$$

The integral can be evaluated using polar coordinates, that is, $y = r\cos\theta$, $z = r\sin\theta$. As in Fig. B1(c), the set H is then

$$H \ : \ r_0 \le r \le r(\theta), \quad \theta_0 \le \theta \le \frac{\pi}{4}, \tag{B3}$$

where

$$\begin{aligned} r_0 &= \sqrt{-\cos t}, \\ r(\theta) &= \sec\theta\, \cos\frac{t}{2}, \\ r_0 \cos\theta_0 &= \cos\frac{t}{2}. \end{aligned} \tag{B4}$$

Eq. (B2) becomes

$$\begin{aligned} \text{Area of H}' &= \sqrt{2} \int_{\theta_0}^{\pi/4} \int_{r_0}^{r(\theta)} \frac{r \sin(t/2)\, \sin t}{(1 + 2r^2 + \cos t)^{3/2}}\, dr\, d\theta \\ &= \frac{\sin t}{2} \left(\theta - \tan\frac{t}{2}\, \sin^{-1}\left(\frac{\sin\theta}{\sqrt{2}}\right)\right)\Bigg|_{\theta=\theta_0}^{\theta=\pi/4}. \end{aligned} \tag{B5}$$

Each of the eight curved triangles in $\mathbb{L}_w(t)$ consists of six regions congruent to H', so the area of $\mathbb{L}_w(t)$ (Fig. 8e) is 48 times the area in eq. (B5).

APPENDIX C: TRIANGLE INEQUALITY FOR d

In this appendix we verify eq. (19d) of Section 3.1, the triangle inequality for d.

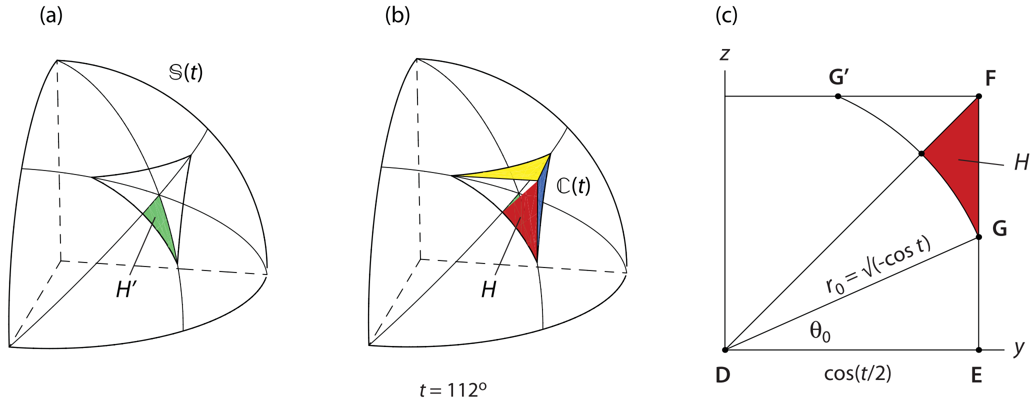

Figure B1. Detail of Fig. 10(f), with $t = 112°$. (a) Set H', making up 1/48[th] of the set $\mathbb{L}_w(t)$ on the sphere $\mathbb{S}(t)$. (b) The corresponding planar set H (red) in $\mathbb{L}_x(t)$ on the cube $\mathbb{C}(t)$. The set H' is the radial projection of H to the sphere. The radial projection is used to parameterize H' and hence to calculate its area. (c) Diagram for writing H in polar coordinates (eqs B4). The cube face $x = \cos(t/2)$ is in the plane of the paper. The points **D**, **E**, **F** are the same as in Fig. 8 and eq. (49). Since the circular arc **GG**′ is on the sphere as well as on the cube face, and since the radius of the sphere is $\sin(t/2)$, the radius of the arc is $r_0 = \sqrt{-\cos t}$ as indicated.

For quaternions q_1 and q_2, we let $\theta(q_1, q_2)$ be the angle between them in $\mathbb{R}^4$. Thus if both q_1 and q_2 are unit quaternions, then $\cos(\theta(q_1, q_2)) = q_1 \cdot q_2$. (The notation $(\cdot)$ is the standard inner product in $\mathbb{R}^4$, not quaternion multiplication.)

If q_1 and q_2 are unit quaternions for rotations U_1 and U_2, and if $q_2 q_1^{-1} = (w, x, y, z)$, then

$$\begin{aligned}\cos\frac{d(U_1, U_2)}{2} &= |w| \\ &= \left|q_2 q_1^{-1} \cdot (1, 0, 0, 0)\right| \\ &= |\, q_2 \cdot q_1| \\ &= |\cos(\theta(q_2, q_1))|\,, \end{aligned} \tag{C1}$$

where the third equation uses the invariance of the inner product under multiplication by a unit quaternion. Hence

$$d(U_1, U_2) = 2\,\theta(q_1, q_2), \tag{C2}$$

where $q_1 \cdot q_2 \geq 0$—as can always be arranged, by changing a sign of q_1 or q_2.

The triangle inequality is known to hold for θ (Berger *et al.* 1984). Therefore, with q_1, q_2, q_3 being quaternions for U_1, U_2, U_3, and with the signs of q_1 and q_3 chosen so that $q_1 \cdot q_2 \geq 0$ and $q_2 \cdot q_3 \geq 0$,

$$\begin{aligned} d(U_1, U_2) + d(U_2, U_3) &= 2\,\theta(q_1, q_2) + 2\,\theta(q_2, q_3) \\ &\geq 2\,\theta(q_1, q_3) \\ &\geq d(U_1, U_3). \end{aligned} \tag{C3}$$

The last line is an inequality rather than an equality, since $q_1 \cdot q_3$ might be negative.

APPENDIX D: TRIANGLE INEQUALITY FOR ξ_0

In this appendix we verify eq. (27c) of Section 3.3, the triangle inequality for ξ_0.

For frames U_1, U_2, U_3, we choose frames $V_1, V_3 \in \mathbb{B}_w$ so that $\overline{V_1} = \overline{U_2^{-1} U_1}$ and $\overline{V_3} = \overline{U_2^{-1} U_3}$. Then

$$\begin{aligned} \xi_0(U_1, U_2) + \xi_0(U_2, U_3) &= \xi_0(U_2^{-1} U_1, I) + \xi_0(I, U_2^{-1} U_3) \\ &= \xi_0(V_1, I) + \xi_0(I, V_3) \\ &= d(V_1, I) + d(I, V_3) \\ &\geq d(V_1, V_3) \\ &\geq \xi_0(V_1, V_3) \\ &= \xi_0(U_2^{-1} U_1, U_2^{-1} U_3) \\ &= \xi_0(U_1, U_3). \end{aligned} \tag{D1}$$

The corresponding fact for principal axis triples then follows from eq. (31):

$$\begin{aligned} \xi_0(\overline{U_1}, \overline{U_2}) + \xi_0(\overline{U_2}, \overline{U_3}) &= \xi_0(U_1, U_2) + \xi_0(U_2, U_3) \\ &\geq \xi_0(U_1, U_3) \\ &= \xi_0(\overline{U_1}, \overline{U_3}). \end{aligned} \tag{D2}$$

APPENDIX E: SAMPLE CALCULATION OF ξ_0

In this appendix we calculate the separation ξ_0 between two earthquakes in New Guinea. These same events were also used as an illustration by Kagan (1991). The first was on 1976 January 6. Its moment tensor, from the GCMT catalog (Dziewonski *et al.* 1981) but with a scale factor omitted, is

$$M_1 = \begin{pmatrix} -0.412 & 0.398 & -1.239 \\ 0.398 & 0.084 & 1.058 \\ -1.239 & 1.058 & 0.328 \end{pmatrix}.$$

We calculate the eigenvectors of M_1, then normalize them and order them according to decreasing size of the eigenvalues. If necessary, the direction of one vector should be reversed to make the triple of vectors right handed. The three vectors become the columns of the matrix

$$U_1 = \begin{pmatrix} 0.404 & 0.642 & 0.651 \\ -0.456 & 0.758 & -0.465 \\ -0.793 & -0.109 & 0.600 \end{pmatrix},$$

which is then one of the four eigenframes associated with M_1. We have displayed numbers to three decimal places, but we retain all places to full machine precision during calculations.

The second event was on 1980 September 26. Its eigenframe U_2 is found as was done for U_1. Then

$$U_1^{-1}U_2 = U_1^T U_2 = \begin{pmatrix} 0.187 & 0.879 & -0.439 \\ -0.651 & -0.224 & -0.725 \\ -0.735 & 0.421 & 0.531 \end{pmatrix}.$$

A quaternion for $U_1^{-1}U_2$, from eqs (F1a) of Appendix F, is

$(0.611,\ 0.469,\ 0.121,\ -0.626)$.

Of the four entries, the largest in absolute value is $z = -0.626$. From eq. (34), the principal axis angle is therefore

$$\xi_0 = 2\cos^{-1}|z| = 102.5^\circ. \tag{E1}$$

The two earthquakes were spatially close. Had they not been close, the principal axis angle would not have been meaningful. Since a sphere is not parallelizable, there is no reasonable way to decide when frames at two widely separated points on Earth should be considered the 'same'. (One can of course regard Earth as a subset of $\mathbb{R}^3$ and define the frames to be the same if one frame is the translate of the other in $\mathbb{R}^3$, but we cannot envision any seismological application where this point of view would be useful.)

Other approaches

A slight variation would be to calculate with quaternions rather than matrices. Eqs (F1) can be used to find unit quaternions q_1 and q_2 for U_1 and U_2. Then the quaternion for $U_1^{-1}U_2$ is $\overline{q}_1 q_2$, which again gives eq. (E1).

Other approaches are also possible. For example, instead of starting with the moment tensors, one can compute unit vectors $\mathbf{t}$ and $\mathbf{p}$ in the T and P directions for each event, using the catalog values of plunge and azimuth for those directions. Ideally, the frame for an event is the matrix U with column vectors $\mathbf{t}$, $\mathbf{b} = \mathbf{p} \times \mathbf{t}$, and $\mathbf{p}$. The vectors $\mathbf{t}$ and $\mathbf{p}$, however, are apt to be not quite orthogonal—due to rounding, for example—and so U, though right-handed, will not be exactly orthogonal. Calculation of the quaternion for U becomes less straightforward, as explained in Appendix F.

There are various ways to orthogonalize U, that is, to replace it with a nearby matrix U' that is orthogonal. A naive but adequate method is to use the matrix U' whose columns are

$$\mathbf{t}' = \mathbf{t},\ \ \mathbf{b}' = \frac{\mathbf{b}}{\|\mathbf{b}\|},\ \ \mathbf{p}' = \frac{\mathbf{t} \times \mathbf{b}}{\|\mathbf{t} \times \mathbf{b}\|}.$$

A slightly better result—that is, closer to U—is obtained by using the appropriate one of q_w, q_x, q_y, q_z from eqs (F1) as a (non-normalized) quaternion for U. Once normalized, the quaternion determines U' by eq. (7). A still better result—in fact theoretically best—is obtained through singular value decomposition (Szabo 2000).

APPENDIX F: QUATERNION FROM ROTATION

A unit quaternion q for a rotation matrix $U = (u_{ij})$ can be computed directly, avoiding eq. (2). For most rotations U, any of the functions q_w, q_x, q_y, q_z in eqs (F1) will give a satisfactory quaternion for U. When dealing with real data, however, for which U may be only approximately orthogonal, the function q_w (respectively, q_x, q_y, q_z) should be avoided when the trace of U (UX_π, UY_π, UZ_π) is close to -1. The trace test becomes less relevant if U has been orthogonalized.

$$\begin{aligned} q_w &= \left(w,\ \frac{u_{32}-u_{23}}{4w},\ \frac{u_{13}-u_{31}}{4w},\ \frac{u_{21}-u_{12}}{4w}\right), \\ w &= \frac{1}{2}(1+u_{11}+u_{22}+u_{33})^{1/2}, \end{aligned} \tag{F1a}$$

$$\begin{aligned} q_x &= \left(\frac{u_{32}-u_{23}}{4x},\ x,\ \frac{u_{12}+u_{21}}{4x},\ \frac{u_{13}+u_{31}}{4x}\right), \\ x &= \frac{1}{2}(1+u_{11}-u_{22}-u_{33})^{1/2}, \end{aligned} \tag{F1b}$$

$$\begin{aligned} q_y &= \left(\frac{u_{13}-u_{31}}{4y},\ \frac{u_{12}+u_{21}}{4y},\ y,\ \frac{u_{23}+u_{32}}{4y}\right), \\ y &= \frac{1}{2}(1-u_{11}+u_{22}-u_{33})^{1/2}, \end{aligned} \tag{F1c}$$

$$\begin{aligned} q_z &= \left(\frac{u_{21}-u_{12}}{4z},\ \frac{u_{13}+u_{31}}{4z},\ \frac{u_{23}+u_{32}}{4z},\ z\right), \\ z &= \frac{1}{2}(1-u_{11}-u_{22}+u_{33})^{1/2}. \end{aligned} \tag{F1d}$$

To verify eqs (F1), first note that

$$\begin{aligned} w &= \frac{1}{2}(1+\mathrm{trace}\,U)^{1/2}, \\ x &= \frac{1}{2}(1+\mathrm{trace}\,UX_\pi)^{1/2}, \quad y = \text{etc.} \end{aligned} \tag{F2}$$

Next, for any 3×3 matrix $U = (u_{ij})$, define the vector $\mathbf{K}$, which determines the skew symmetric part of U, by

$$\mathbf{K}(U) = (u_{32}-u_{23},\ u_{13}-u_{31},\ u_{21}-u_{12}). \tag{F3}$$

If U is a rotation matrix with unit quaternion $q = (w, x, y, z)$, then, from eq. (7),

$$\begin{aligned} 4w^2 &= 1 + \mathrm{trace}\,U, \\ 4w\,(x, y, z) &= \mathbf{K}(U). \end{aligned} \tag{F4}$$

If trace $(U) \neq -1$, then eqs (F4), together with a choice of sign for w, determine w, x, y, z, and hence q. The quaternion q_w in eqs (F1a) is the solution (w, x, y, z) of eqs (F4) with $w > 0$.

Applying eqs (F4) to UX_π, rather than to U, gives

$$\begin{aligned} 4x^2 &= 1 + \mathrm{trace}\,UX_\pi, \\ 4x\,(-w, -z, y) &= \mathbf{K}(UX_\pi), \end{aligned} \tag{F5}$$

where $(w,\ x,\ y,\ z)$ is still the quaternion for U (not UX_π). The quaternion q_x in eqs (F1b) is the solution of eqs (F5) with $x > 0$. Eqs (F1c) and (F1d) are verified similarly.

To see why one of q_w, q_x, q_y, q_z (eqs F1) is sometimes preferable to the others, we find from eqs (F1a) that

$$\frac{\partial q_w}{\partial u_{11}} = \frac{\overline{q_w}}{8w^2}, \tag{F6}$$

where the bar denotes quaternion conjugate (Section 2.2). The partial derivatives of q_w with respect to the other u_{ij} are like $1/(8w^2)$ or $1/(4w)$. A perturbation in U therefore can produce a perturbation in q_w that is larger by a factor of about

$$f_w(U) = \frac{1}{8w^2} = \frac{1}{2(1+\mathrm{trace}\,U)} = \frac{1}{8\cos^2(\xi/2)}, \tag{F7}$$

where ξ is the rotation angle of U. The parameter f_w is therefore a measure of the numerical instability of q_w. For rotation angle $\xi = 179°$, the parameter f_w is about 1600, whereas for $\xi = 120°$ it is 0.5. When dealing with real data, one clearly should avoid using q_w when trace$(U) \approx -1$ or, equivalently, when ξ is large. (Our talk of rotation angle is somewhat loose, since the matrices in question might not be precisely rotations.)

For $q_x, q_y, q_z,$ the instability factors are, respectively,

$$f_x(U) = \frac{1}{8x^2} = \frac{1}{2(1 + \operatorname{trace} UX_\pi)} = \frac{1}{8\cos^2(\xi_x/2)},$$
$$f_y(U) = \frac{1}{8y^2} = \frac{1}{2(1 + \operatorname{trace} UY_\pi)} = \frac{1}{8\cos^2(\xi_y/2)}, \qquad \text{(F8)}$$
$$f_z(U) = \frac{1}{8z^2} = \frac{1}{2(1 + \operatorname{trace} UZ_\pi)} = \frac{1}{8\cos^2(\xi_z/2)},$$

where as in Section 3.7, the parameters ξ_x, ξ_y, ξ_z are the rotation angles for UX_π, UY_π, UZ_π. For a given U the best choice q_0 among q_w, q_x, q_y, q_z is therefore determined by which of U, UX_π, UY_π, UZ_π has the largest trace or, equivalently, by which has the smallest rotation angle. The instability factor for q_0 is therefore

$$f_0(U) = \frac{1}{8\cos^2(\xi_0/2)}, \qquad \text{(F9)}$$

where ξ_0 is the principal axis angle. Thus the level surfaces for f_0 are the same as those for ξ_0 (Fig. 10). The parameter f_0 varies from 0 to 0.5. The minimum occurs at the identity and at the 180° rotations about the coordinate axes, and the maximum occurs at the 120° rotations about axes in the $(\pm 1, \pm 1, \pm 1)$ directions.

In practice, if U has been orthogonalized, then we usually need not worry about q_0; any of q_w, q_x, q_y, q_z will work, unless the relevant trace happens to be extremely close to -1. The problems arise when U is only approximately orthogonal; then the choice among q_w, q_x, q_y, q_z can matter.

Eqs (F1) are found in, for example, Hanson (2006). Bar-Itzhack (2000) describes other treatments of this material.

Mathematical perspective

To avoid having to treat multiple sign possibilities, we define the set $\mathbb{U}^+ \subset \mathbb{U}$ to consist of the rotations that have a unit quaternion $q = (w, x, y, z)$ for which all of w, x, y, z are positive. From a purely mathematical point of view, all of the functions q_w, q_x, q_y, q_z coincide on $\mathbb{U}^+$. Their directional derivatives at $U \in \mathbb{U}^+$ must also agree, if the directions are tangent to $\mathbb{U}$. But the directional derivatives at U may not agree in directions transverse to $\mathbb{U}$, since the functions themselves in general do not agree outside $\mathbb{U}$. This is how, for example, $\partial q_w/\partial u_{11} \neq \partial q_x/\partial u_{11}$, even though q_w and q_x agree on $\mathbb{U}^+$. And this is why the choice of q_w, q_x, q_y, q_z can matter.